

Tight Bounds on Uniform-Challenge Reductions from Sigma Protocols

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Abstract

Sigma protocols are fundamental cryptographic tools, serving as the foundation of many practical schemes—most notably, the *Schnorr identification and signature schemes*. To prove the security of Sigma protocols, one typically reduces breaking a Sigma protocol to solving a presumed hard problem (e.g., computing the discrete logarithm in a certain group). In many settings, however, these reductions are not tight: given an adversary that breaks a Sigma protocol with probability ε , the reduction only yields an adversary for the underlying problem with probability ε^2 . This quadratic loss affects efficiency, as it forces choosing larger security parameters to reach a target security level.

In this work, we show that this quadratic loss is *inherent* for two natural classes of reductions. For interactive protocols, we prove it for *uniform-challenge, black-box* reductions, which query the adversary using uniformly sampled challenges. For non-interactive protocols (i.e., in the random-oracle model), we prove it for *weakly programmable, black-box* reductions, which answer the adversary’s oracle queries with uniformly sampled outputs. Applying our bounds to the reductions from Schnorr identification and signatures to discrete logarithm yields lower bounds that match known positive results—namely, the classical worst-case reduction of Pointcheval and Stern (Journal of Cryptology, 2000) and the higher-moment reduction of Rotem and Segev (Journal of Cryptology, 2024).

Our approach reduces the analysis of such reductions to the values of simple *hitting games*—combinatorial games that we introduce. Bounding these games is our main technical contribution, and we believe these bounds can enable more modular proofs of related results.

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1 Introduction

Sigma protocols are three-message, public-coin proof systems; on common input (i.e., instance) x , the prover talks first by sending a so-called *committing message*, the verifier replies with a random uniform *challenge*, and the prover finalizes the protocol by issuing a *response*. The verifier then accepts or rejects, where it accepts if and only if the prover knows a valid witness for x being in the language of interest. Sigma protocols are among the most useful cryptographic tools, serving as the basis for many simple and practical schemes. For instance, they can be readily employed as identification schemes: a party’s identity is tied to a pair $(x, w) \in \mathcal{R}$, for some NP-relation $\mathcal{R}$, and then the party can prove its identity to an authority by executing the Sigma protocol—acting as the prover while the authority plays the verifier. Numerous identification schemes have been built on Sigma protocols (e.g., [GQ88; Sch91; Bet88; MS90; BM91; Gir91; Oka93; FS87]). Of these, the Schnorr identification scheme [Sch91] is the most widely adopted; it relies on the hardness of the discrete-logarithm problem (hereafter discrete-log) in a (public) cyclic group.

To prove the security of Sigma protocols, we typically reduce the problem of breaking them (convincing the verifier of a false statement, or without knowledge of a witness) to solving computational problems that are believed to be hard (e.g., discrete-log). In many settings, however, these security reductions are not tight. Given an adversary that breaks the Sigma protocol with probability ε , the reduction constructs an adversary that breaks the underlying hard problem with probability ε^2 . When considering concrete security parameters, this quadratic security loss adversely affects efficiency: the protocol must be executed with larger security parameters to preserve the provable security guarantee, resulting in longer computation times and, more significantly, increased communication complexity.¹

Sigma protocols can be made non-interactive using the Fiat-Shamir paradigm [FS87], where the prover computes the verifier’s challenge via a cryptographic hash function (e.g., one from the SHA family),² and, to argue security, the hash function is modeled as a random oracle [BR93]. Even in this idealized model, however, the security analysis suffers from the same—and even worse—security loss issues. Namely, given an adversary that breaks the non-interactive Sigma protocol with probability ε , the reduction Pointcheval and Stern [PS00] constructs an adversary that breaks the underlying hard problem with probability ε^2/b where b denotes a bound on the number of random oracle queries the adversary makes. This state of affairs raises the question of whether this security loss is inherent.

1.1 Our Results

We show that for a restricted yet natural family of reductions, the above quadratic security loss is inevitable for both interactive and non-interactive Sigma protocols. In both cases, we assume the Sigma protocol is honest-verifier zero knowledge.

Interactive Sigma protocols. The following bounds are for interactive Sigma protocols and thus readily extend to *Schnorr identification schemes*. Our bounds are for *uniform-challenge black-box reductions*: black-box reductions that sample the emulated challenges uniformly. These terms

¹In practice, parameters are often set optimistically based on the best-known attacks. This approach, however, undermines our confidence in the scheme’s security, as it effectively bases its security on the scheme itself rather than on a well-established hard problem, like discrete-log or factoring.

²This approach gives rise to *signatures* by embedding the message to be signed in the argument of the hash.

are explained in detail once we introduce the following terminology: a *game* G is just an interactive algorithm. A *player/algorithm* A ε -breaks (the security of) G if G accepts, i.e., outputs 1, in the interaction with A with probability (at least) ε . For instance, G might send a random group element X to A and accept if A finds the discrete-log. Game G is (t, ε) -secure if no t -time algorithm ε -breaks G . *Black-box reductions* between (the security of) two games are defined as follows:

Definition 1.1 (Black-box reductions, informal). *Algorithm R is a black-box reduction from the (t_2, ε_2) -security of game G_2 to the (t_1, ε_1) -security of game G_1 , if given any A that breaks the ε_2 -security of G_2 , algorithm R^A breaks the ε_1 -security of G_1 . In addition, when charging t_2 steps for each A -call, R^A runs in time t_1 .*

In particular, if A violates the (t_2, ε_2) -security of G_2 , then R^A violates the (t_1, ε_1) -security of G_1 . (Note, however, that R^A should break the ε_1 -security of G_1 *regardless* of A 's running time.³) In this work, we focus on reductions from games induced by Sigma protocols. For a Sigma protocol Σ defined by the prover's P committing message, the verifier's V challenge, and P 's response, the game G_Σ is defined as follows: the challenger acts as the verifier V of Σ , on some input x , and accepts if V does. Our first result is an emulation of *uniform-challenge* reductions from the security of G_Σ .

Definition 1.2 (Uniform-challenge reductions, informal). *A black-box reduction from the security of G_Σ is uniform-challenge, if it samples the challenges it sends to A uniformly and independently (in the challenge space specified by Σ).*

Theorem 1.3 (Emulating uniform-challenge reductions, informal). *Let R be a uniform-challenge reduction from the $(t_\Sigma, \varepsilon_\Sigma)$ -security of G_Σ to the (t_G, ε_G) -security of a game G . Then, there exists an adversary A that ε_Σ -breaks the security of G_Σ and a (non oracle-aided) algorithm $\widehat{R}$ such that following holds:*

1. *The execution of $(\widehat{R}, G)$ is DH-close to that of (R^A, G) (in statistical distance), for DH being the probability that R^A asks A two (different) challenges on which A returns valid answers.*
2. *The running time of $\widehat{R}$ is at most that of R in R^A (i.e., its running time without incurring any overhead for the oracle calls).⁴*

In particular, the probability that $\widehat{R}$ wins G is DH-close to the probability that R^A wins, and it does so without spending more time than R^A does (counting the running time of each A -call as zero). In order to bound DH, we note that A returns valid answers with probability ε_Σ (by design), and R makes at most t_G/t_Σ queries. Therefore, by simple union bound, $\text{DH} \leq (t_G/t_\Sigma)^2 \cdot \varepsilon_\Sigma^2$. Hence, Theorem 1.3 yields the bound below for uniform-challenge reductions. We emphasize that both the above and the following bounds apply to *any* game G , not only to non-interactive games (in contrast to some other results [FJS19]).

Corollary 1.4 (Bounds on uniform-challenge reductions, informal). *For any uniform-challenge reduction from the $(t_\Sigma, \varepsilon_\Sigma)$ -security of G_Σ to the (t_G, ε_G) -security of a game G : if G is $(t_G, t_G^2/s)$ -secure, then $\varepsilon_\Sigma \in \Omega(t_\Sigma/\sqrt{s})$.⁵*

³This is the standard definition of black-box reduction, see [RTV04] for discussion.

⁴More formally, there exists a coupling between the coins of $\widehat{R}$ and that of R and A , for which this bound on the running time of $\widehat{R}$ holds.

⁵Here and in the following bounds, we also assume that ε_G is not too close to the assumed lower bound, t_G^2/s . See Footnote 9 and section 4.1.2 for further discussions.

In particular, for an s -size group where the discrete-log game is $(t_G, O(t_G^2/s))$ -hard, for any t_G , as we believe is the case for most elliptic curve groups we use in practice, the (uniform-challenge) reduction of Pointcheval and Stern [PS00] from the security of the Schnorr identification scheme to the hardness of discrete-log is tight. Put differently, a uniform-challenge reduction that improves upon the reduction of [PS00] yields that discrete-log in the underlying group is *not* $(t_G, O(t_G^2/s))$ -secure.

Non-interactive Sigma protocols. The above bounds extend (with better parameters) to the non-interactive variant of Sigma protocols, and thus to Schnorr signatures. In this case, the family of reductions we consider are what we call *weakly-programming black-box reductions*, which adapt the previous terminology to oracle-aided non-interactive games. Let G be an oracle-aided algorithm and $\mathcal{H}$ a function family. An algorithm A is said to ε -break $G^\mathcal{H}$ if, for $h \xleftarrow{R} \mathcal{H}$, the algorithm G^h accepts in its interaction with A with probability ε .⁶ Game $G^\mathcal{H}$ is said to be (t, ε, b) -secure, if no b -query, t -time algorithm ε -breaks G .

An interactive game G in which the challenger sends uniformly sampled messages, such as a Sigma protocol game G_Σ , naturally gives rise to a non-interactive Fiat–Shamir variant $G^\mathcal{H}$. In this variant, the challenger’s messages are generated by the player by applying an oracle function $h \in \mathcal{H}$ to the transcript of messages that the challenger would have sent up to that point.⁷

Definition 1.5 (Weakly-programming, black-box reductions, informal). *A black-box reduction from the security of $G_\Sigma^\mathcal{H}$ is weakly programming if when emulating an adversary A that breaks the security of $G_\Sigma^\mathcal{H}$, the reduction answer its oracle calls uniformly: on input x it either returns the answer it gave previously on x , or returns a uniform element in the domain, i.e., “it resamples the oracle answer”.*

Note that the above is the natural adaptation of uniform-challenge reductions to the non-interactive oracle-aided world; since the player/adversary samples the challenger’s messages using the oracle h , a weakly programming reduction can effectively call it on uniformly sampled challenges. Both Theorem 1.3 and Corollary 1.4 extend to the non-interactive setting; however, for brevity, we only state the analogue of Corollary 1.4 here.

Corollary 1.6 (Bounds on weakly-programming reductions, informal). *For any weakly-programming reduction from the $(t_\Sigma, \varepsilon_\Sigma, b)$ -security of $G_\Sigma^\mathcal{H}$ to the (t_G, ε_G) -security of a game $G^\mathcal{H}$: if $G^\mathcal{H}$ is $(t_G, t_G^2/s)$ -secure then $\varepsilon_\Sigma \in \Omega(\sqrt{b} \cdot t_\Sigma / \sqrt{s})$ if $(t_G/t_\Sigma)\varepsilon_\Sigma < 1$ and $\varepsilon_\Sigma \in \Omega(b \cdot t_G t_\Sigma / s)$ otherwise.*

That is, this is essentially the same bound as in the interactive case, but it holds for much larger values of ε_Σ , matching the known upper bound for reductions from the security of non-interactive Sigma protocols to the hardness of discrete-log Pointcheval and Stern [PS00] and Seurin [Seu12].

1.1.1 Higher-Moment Reductions

In a recent work, Rotem and Segev [RS24] presented reductions from the security of Schnorr identification and signature schemes to the d -moment hardness of discrete-log. The d -moment

⁶For instance, for a cyclic group $\mathcal{G}$ generated by some $G \in \mathcal{G}$, G might send $X = x \cdot G$ for a uniform x to A , who replies with $A \leftarrow a \cdot G, z \leftarrow a + h(X, A) \cdot x$. Then G accepts if $z \cdot G = A + h(X, A) \cdot X$. Note that this corresponds to the classic Schnorr signature scheme game.

⁷At the end of the game, the player sends all its messages and the challenger samples the relevant challenge messages using h , then verifies correctness.

running time of an algorithm is defined by $E[T^d]$, for T denoting its running time in a random execution. Rotem and Segev [RS24] presented a reduction from the security of Schnorr identification and signature schemes to the d -moment hardness of discrete-log: no algorithm with d -moment running time t_{DL} can find discrete log with probability better than t_{DL}/s , for s being the size of the underlying group. While this assumption is false for $d = 1$, it is plausible for $d \geq 2$, and in particular for $d = 2$, it was recently shown to hold unconditionally in the generic group model by Segev, Sharabi, and Yosev [SSY23]. Rotem and Segev [RS24] overcome the classical quadratic loss in reductions to worst-case discrete-log hardness: an adversary that breaks the Schnorr identification scheme with probability ε can be transformed into a 2-moment adversary for discrete-log that succeeds with probability $\varepsilon^{3/2} \in \omega(\varepsilon^2)$. This raises the question whether such reductions can be pushed further, or even be made lossless—the success probability of the reduction in breaking discrete-log is the same as that of the adversary in breaking the Sigma protocol. We prove that is not the case for the type of reductions under consideration, showing that the reductions of [RS24] are tight for this class of reductions.

Interactive Sigma protocols. A game G is (t_G, ε, d) -secure if no algorithm with d -moment running time t_G ε -breaks G . To give an analogue of Corollary 1.4, we need to bound DH_d , the probability that a reduction with d -moment running time t_G asks the adversary A two valid challenges. We prove the tight bound $DH_d \leq ((t_G^{1/d}/t_\Sigma)\varepsilon_\Sigma)^{2-1/d}$. Thus, by Theorem 1.3, we deduce the following result.

Corollary 1.7 (Bounds on uniform-challenge, higher-moment, reductions, informal). *For $d \geq 2$ and a uniform-challenge reduction from the $(t_\Sigma, \varepsilon_\Sigma, d)$ -security of G_Σ to the (t_G, ε_G) -security of a game G : if G is $(t_G, t_G/s, d)$ -secure then $\varepsilon_\Sigma^{2-1/d} \in \Omega(t_\Sigma^d/s)$.*

In particular, for a group of size s that is $(t, O(t/s), d)$ -secure—as is the case for generic groups when $d = 2$ [SSY23]—the uniform-challenge reduction of [RS24] from the security of the Schnorr identification scheme to the d -moment hardness of discrete logarithm is tight.

Non-interactive Sigma protocols. Higher-moment security naturally extends to the non-interactive setting: a game G^H is (t_G, ε, d, b) -secure if no b -query algorithm with d -moment running time t_G ε -breaks it. We prove the following bound.

Corollary 1.8 (Bounds on weakly-programmable, higher-moment, reductions). *For $d \geq 2$ and a weakly-programming reduction from the $(t_\Sigma, \varepsilon_\Sigma, d, b)$ -security of G_Σ to the (t_G, ε_G) -security of a game G : if G is $(t_G, t_G/s, d)$ -secure then $\varepsilon_\Sigma^{2-1/d} \in \Omega(b \cdot t_\Sigma^d/s)$ if $(t_G^{1/d}/t_\Sigma)\varepsilon_\Sigma < 1$ and $\varepsilon_\Sigma^d \in \Omega(b \cdot t_\Sigma^d/s)$ otherwise.*

In particular, for an s -size group that is $(t, O(t/s), d, b)$ -secure, the above bound matches the weakly-programmable reduction of Rotem and Segev [RS24] from the security of the Schnorr signature scheme to the d -moment hardness of discrete-log.

1.1.2 Beyond Uniform-Challenge Reductions

We now discuss extensions and limitations of our results, beginning with non-uniform challenge distributions. On the positive side, when the zero-knowledge simulator of the Sigma protocol works for *any* challenge (as in Schnorr-type Sigma protocols), our bounds extend to reductions that query

the adversary under an arbitrary but *fixed* challenge distribution (not necessarily uniform). On the negative side, our techniques do not apply to *unconstrained* black-box reductions that place no conditions on the challenge distribution. There is good reason for this limitation: without such constraints there is always a trivial, tight black-box reduction from the security of a Sigma protocol to itself.

When focusing on reductions to non-interactive games such as discrete logarithm (thereby avoiding trivial self-reductions), it is plausible that stronger impossibility results may hold—potentially ruling out *any* black-box reduction, including those without constraints on the challenge distribution. Indeed, for reductions to discrete log, prior works exclude unconstrained black-box reductions, but only under additional assumptions on the group (e.g., the *one-more discrete-log* assumption) and/or by restricting the reduction to algebraic/generic models (see Section 1.3). Ruling out arbitrary black-box and, even more so, non-black-box reductions to discrete log remains a central open problem.

Finally, as noted in Section 1.3.2, there exist *tight*⁸ black-box reductions from the security of Schnorr identification and signatures to other (less widely studied) hardness assumptions that use *non-uniform* challenge distributions. These lie outside our framework and are therefore not ruled out by our bounds. Whether non-uniform challenge distributions can likewise yield tight black-box reductions to discrete-log hardness remains an open question.

1.2 Our Technique

In what follows, we outline our proof approach, focusing for concreteness on interactive Σ -protocols. Let $\Sigma = (P, V)$ be a Σ -protocol with common input x and prover’s witness w . The prover sends a commitment a , the verifier samples a challenge e , and the prover replies with a response z ; the transcript is $\tau = (x, a, e, z)$, and verification is $V(x, a, e, z) = 1$ if and only if the transcript is valid. Further, let R be uniform-challenge reduction from the $(t_\Sigma, \varepsilon_\Sigma)$ -security of G_Σ to the (t_G, ε_G) -security of a game G . The (inefficient) adversary A we use for proving the bound is straightforward:

Algorithm 1.9. [Adversary A]

On input-query x :

1. Use brute force to find a witness w for x .
2. Sample a random ε_Σ -dense subset $\mathcal{G}$ from the challenge space $\mathcal{E}$ of Σ .
3. Output the first message of the prover according to (x, w) .

On challenge-query e :

If $e \notin \mathcal{E}$, return $\perp$.

Else, use (x, w, a) to compute a valid z , and output it.

It is clear that A breaks G_Σ with probability ε_Σ . (We also note that A is inefficient, but since the reduction R is black-box, it should perform as expected given oracle access to A . See Footnote 3)

⁸That is, $\varepsilon_R \approx \varepsilon_A$, where ε_R and ε_A denote the success probabilities of the reduction and the attacker, respectively, and the reduction’s running time is essentially that of the attacker.

The accompanying emulator $\widehat{R}$ for R^A acts as follows:

Algorithm 1.10. [Emulator $\widehat{R}$] Start emulating R , each time it calls A on input x :

1. Use the zero-knowledge to Simulator of Σ to find accepting transcript (a, e, z) for x .
2. Send the committing message “ a ” back to R as A ’s answer.

Whenever R makes a (uniform) challenge query to A :

With probability $1 - \varepsilon_\Sigma$: set the challenge to $e' \xleftarrow{R} \mathcal{E} \setminus \{e\}$, and A ’s answer to $\perp$.

Else, set the challenge to e , and A ’s answer to z .

It is not hard to see that $\widehat{R}$ perfectly emulates R^A until R *samples two different queries on which A returns non- $\perp$ answers*. Thus, we have established Theorem 1.3.

1.2.1 The Double-Hitting Game

To bound the probability that R samples two distinct queries on which A returns a non- $\perp$ answer, and thereby prove Corollaries 1.4 and 1.7, we introduce the *double-hitting game*. Let Ber_ε denote a Bernoulli trial that returns 1 with probability ε .

Game 1.11. [Double hitting]

Parameter: $\varepsilon \in [0, 1]$.

Parties: Challenger C and player P .

The game:

1. P (adaptively) queries C , and gets $c \xleftarrow{R} \text{Ber}_\varepsilon$.
2. P **wins** if the value 1 is returned for at least *two* queries.

It is easy to see that the probability that R samples two distinct queries on which A returns a non- $\perp$ answer is upper-bounded by the success probability of a q -query player in the double-hitting game (Game 1.11) with parameter $\varepsilon \leftarrow \varepsilon_\Sigma$, where q is the query complexity of R . Hence, any upper bound on the value of this game (i.e., the winning probability of a q -query player) yields the same bound on the loss incurred when emulating uniform-query reductions.

Since the query complexity of R is at most t_G/t_Σ (each oracle call costs t_Σ steps), to make Theorem 1.3 useful it suffices to bound the winning probability of a $(q \leftarrow t_G/t_\Sigma)$ -query player in the double-hitting game.

Worst-case query complexity. For a worst-case q -query player (equivalently, a worst-case running-time reduction R), a union bound gives a winning probability at most $q^2 \varepsilon_\Sigma^2$ (there are at most q^2 ordered query pairs, each hitting with probability ε_Σ). Hence, by the emulation argument, the efficient oracle-free algorithm $\widehat{R}$ breaks G with probability at least $\varepsilon_G - q^2 \varepsilon_\Sigma^2$. If G is

$(t_G, t_G^2/s)$ -secure over a group of size s and $\varepsilon_G \geq 2t_G^2/s$,⁹ then $q^2\varepsilon_\Sigma^2 \geq \varepsilon_G - t_G^2/s \geq t_G^2/s$. Using $q = t_G/t_\Sigma$ yields $\varepsilon_\Sigma^2 \geq t_\Sigma^2/s$, which yields Corollary 1.4.

Higher-moments query complexity players. For higher-moment query-complexity players, which corresponds to higher-moment running-time R , the naive bound $q^2\varepsilon_\Sigma^2$ is far too weak for proving Corollary 1.7. To see why, we briefly discuss a player P_0 for Game 1.11 (implicitly defined in [RS24]): P_0 makes a single query to C . If unsuccessful, it stops. Otherwise, it makes additional $\varepsilon_\Sigma^{-1/d}$ queries. It is easy to see that P_0 has d^{th} -moment query complexity $O(1)$, and [RS24] show that its success probability is at least $\varepsilon_\Sigma^{2-1/d}$ ($\varepsilon_\Sigma^{-1/d}$ -far from the naive bound $\varepsilon_\Sigma^2 \cdot \varepsilon_\Sigma^{-2/d}$). To match this attack, and thus proving Corollary 1.7, we prove the following tight bound.

Theorem 1.12 (Bounds for higher-moment double-hitting, informal). *A player with d -moment query complexity q wins Game 1.11 with parameter ε_Σ with probability at most $(q^{1/d}\varepsilon_\Sigma)^{2-1/d}$.*

Proving the above bound, and the analogue bound for its non-interactive variant (discussed below), is the most technically challenging part of this paper.¹⁰

1.2.2 Non-interactive Sigma Protocols

To prove bounds for non-interactive reductions, we adapt the adversary and emulator from the interactive setting.¹¹ In place of the double-hitting game, we introduce the *coins-and-cows game*, which tightly captures the power of weakly programmable reductions. As in the interactive case, the worst-case q -query regime admits an immediate tight upper bound on the winning probability, yielding Corollary 1.6. With additional technical work, we extend the analysis to d -moment query complexity and obtain a matching bound, which yields Corollary 1.8.

1.2.3 Applications to Other Types of Black-Box Reductions

The hitting games described above appear to capture the technical core of several existing lower bounds on black-box reductions from both interactive and non-interactive Sigma protocols. In particular, this seems to hold for the bounds in [PV05; GBL08; Seu12; FJS19] (see Section 1.3 for further discussion). A formal study of this connection could unify much of the prior work in this area and is likely to yield analogous bounds for higher-moment reductions of the type considered here. We leave this important project for future work.

1.3 Related Work

For concreteness, fix a p -order additive cyclic group $\mathcal{G}$ generated by G . Recall that the Schnorr identification scheme (P, V) consists of a three-move protocol that on a group element $X = x \cdot G$ given as common input (i) P sends $A = a \cdot G$ for a random a (ii) V returns a challenge e (iii) P

⁹ Otherwise, the security induced by the reduction critically relies on the spooky assumption that G is $(t, 2t^2/s)$ -secure. For the discrete-log game, for instance, the known t^2/s attack can be easily improved to $2 \cdot t^2/s$ by using, say, atomic operations that provide constant speedup.

¹⁰ Unfortunately, we found no one-line, or even few-line, description that captures the essence of these proofs, but saying that they are obtained via a subtle induction on the maximal number of queries the player does, and on the number of hits it obtains. See details in Section 5.

¹¹ Our adversary construction is also inspired by [Seu12].

responds with $z = ex + a \bmod p$, and V accepts if $z \cdot G = e \cdot X + A$. The Schnorr signatures scheme for public key $X \in \mathcal{G}$, correspond to the non-interactive variant of the Schnorr identification scheme where V 's challenge, e , is generated via a cryptographically secure hash function on input A and the message to be signed. In the following discussion, we focus on Schnorr signatures and identification schemes. We note, however, that some of the discussed results extend to general Schnorr-type Sigma protocols.

1.3.1 Lower Bounds

Starting with Paillier and Vergnaud [PV05], there is an extensive body of work on lower bounds for Schnorr signatures using the so-called *emulation* technique (also known as meta-reductions, or “reduction against the reduction”). In this technique, given a reduction R that interacts with a forger A , the emulation $\widehat{R}$ mimics A 's behavior without using it, sometimes using an additional powerful external oracle (e.g., the *one-more-discrete-log*, OMDL, oracle). Note that our proofs follow exactly this blueprint, where the emulators are using the guaranteed zero-knowledge simulator to emulate A 's answers—see Section 1.2.

Algebraic reductions assuming OMDL. Very loosely, an algorithm is *algebraic* if for any group element Y it produces, the discrete log can be extracted from its random tape.¹² In [PV05; GBL08; Seu12], the following emulation strategy for an *algebraic* reduction R proving the security of Schnorr signatures is considered. For simplicity, assume R succeeds given a single forgery. The emulator $\widehat{R}$ is granted access to a *one-more-discrete-log* oracle OMDL, i.e., it may issue a single discrete-log query. On a pair of group elements (A_0, A_1) , $\widehat{R}$ starts emulating $R^A(A_0)$. When R calls A on a public key X , it returns a forgery $(\alpha \cdot A_1, z)$ with respect to the public key X , for random α , where $z \cdot G = \alpha \cdot A_1 + e \cdot X$ and $e = h(X, \dots)$ is one of the random oracle answers sampled by R . For computing z , $\widehat{R}$ uses the OMDL oracle. If, at the end of the emulation, R outputs the discrete logarithm a_0 of A_0 , then $\widehat{R}$ also extracts the discrete logarithm of A_1 ; by the algebraic property of R , this can be done efficiently using the signature pair together with the known α and x_0 .¹³ Effectively, $\widehat{R}$ finds the discrete log of A_0 and A_1 by making a single query to OMDL. Then, using a hitting-game-style argument, Seurin [Seu12] proves that

$$\varepsilon_{\widehat{R}} \geq \varepsilon_R - \frac{q \cdot \varepsilon_A}{b} \quad (1)$$

for $\varepsilon_{\widehat{R}}$, ε_R and ε_A being the success probability of $\widehat{R}$, R and A in their respective tasks, q is the number of times R invokes A and b is the number of random oracle queries made by A . The loss $q \cdot \varepsilon_A / b$ in the success probability essentially corresponds to the probability that R makes A output two distinct forgeries (A, e, z) and (A, e', z') for $e \neq e'$.

(As noted in Section 1.2, this loss essentially matches the loss in our emulation.) Under the OMDL assumption, $\varepsilon_{\widehat{R}}$ is negligible, and thus Equation (1) (essentially) implies $\varepsilon_A \geq b \varepsilon_R / q$. In our terminology, this coincides with the worst-case bound of Corollary 1.6 (in our setting, $\varepsilon_R \in \Omega(t_R^2/s)$). The same approach extends to reductions from the security of Schnorr signatures to the hardness of OMDL (rather than to discrete log).

¹²In the additive group notation we use here, we can find y so that $Y = y \cdot G$

¹³Since $\widehat{R}$ is algebraic, the emulator can extract u, v such that $X = u \cdot A_0 + v \cdot G$ (as A_0 and G are the only two points fed to $\widehat{R}$ with unknown discrete log relation). Then, if $\widehat{R}$ returns a_0 such that $A_0 = a_0 \cdot G$, it is easy to deduce $x \leftarrow ua_0 + v$ such that $X = x \cdot G$, and, using the forgery z , the emulator outputs $a_1 \leftarrow (z - ex)/\alpha$ such that $A_1 = a_1 \cdot G$.

Generic reductions. In a more recent work, Fleischhacker, Jager, and Schröder [FJS19] considered a sub-class of algebraic reductions called *generic reductions*. Intuitively, a reduction is generic if it works over any representation of the group, or, equivalently, if the identifying labels of the group elements are permuted before they are fed to the reduction. [FJS19] showed that for such reductions, it is possible to replace the role of the OMDL oracles used for emulating reductions above with the ability to choose a random group element for which it knows the discrete log. Given this additional power, essentially the same approach used for algebraic reductions yields a similar bound on generic reductions. These bounds readily extend to reduction to any non-interactive computational game.

Single-instance, non-programming reductions. Fischlin and Fleischhacker [FF13] proved an impossibility result for *single-instance, non-programmable* reductions in the random-oracle model. Such a reduction cannot modify the oracle (i.e., it cannot program it) and interacts with the adversary in only a *single* session tied to a single public key. They show that Schnorr signatures *cannot* be proven secure by such reductions.

1.3.2 Upper Bounds (Security Reductions)

Reductions to discrete-log. The reductions of the Schnorr identification and Schnorr signatures to discrete log are a classic result in cryptography. For the Schnorr identification scheme, Pointcheval and Stern [PS00] (via the *splitting lemma*) and Bellare and Palacio [BP02] (via the *reset lemma*) proved that any t -time attacker mounting an impersonation attack with success probability ε can be transformed into a t -time attacker that breaks discrete log with probability ε^2 . Rotem and Segev [RS24] proved that a t -time attacker mounting an impersonation attack with success probability ε can be transformed into a d -moment, running-time t^d attacker that breaks discrete log with probability $\varepsilon^{2-1/d}$.

For signature schemes, Pointcheval and Stern [PS00] (via the *forking lemma*) proved that a t -time, b -query attacker that forges a signature with probability ε can be transformed into a t -time attacker that breaks discrete-log with probability ε^2/b . Rotem and Segev [RS24] proved that an attacker with t -time, b -query that forges a signature with probability ε can be transformed into a d^{th} -moment, running-time t^d attacker that breaks discrete-log with probability $\varepsilon^{2-1/d}/b$. An indication for the importance of the latter reduction was given by Segev, Sharabi, and Yorgev [SSY23], who proved that, in *the generic group model*, and for suitable t^d , a d^{th} -moment, running-time t^d attacker cannot break discrete-log with success probability $\varepsilon^{2-1/d}/b$.

Reductions to other assumptions. Bellare and Dai [BD20] gave a tight black-box reduction from the security of the Schnorr identification and signature schemes to a variant of the OMDL problem. More recently, Cho, Fuchsbaauer, O’Neill, and Sefranek [CFOS25] gave a tight black-box reduction from the Schnorr identification and signature schemes to a falsifiable hardness assumption that they introduced. These reductions circumvent our impossibility barriers by not using uniform challenges, and circumvent the bounds of [FJS19] by reducing to an interactive game for Bellare and Dai [BD20] and not being generic for Cho, Fuchsbaauer, O’Neill, and Sefranek [CFOS25].

Constructions in idealized models. Fuchsbaauer, Plouviez, and Seurin [FPS20] prove that in the algebraic group plus a random oracle model, Schnorr signature schemes are as secure as discrete-log (i.e., no security loss).

Paper Organization

Section 2 contains definitions and background material. Section 3 describes the emulator for uniform-query reductions of interactive and non-interactive Sigma protocols. Section 4 introduces several hitting games and uses the bounds on their values (proven in Section 5) with the results of Section 3 to upper-bound the quality uniform-query reductions. Finally, upper and lower bounds on the hitting games introduced in Section 4 are proven in Section 5.

2 Preliminaries

2.1 Notation

We use calligraphic letters to denote sets, uppercase for random variables, and lowercase for integers and functions. Let $\mathbb{N}$ denote the set of natural numbers. For $n \in \mathbb{N}$, let $[n] := \{1, \dots, n\}$ and $(n) := \{0, \dots, n\}$. For a relation $\mathcal{R}$, let $\mathcal{L}(\mathcal{R})$ denote its underlying language, i.e., $\mathcal{L}(\mathcal{R}) := \{x : \exists w : (x, w) \in \mathcal{R}\}$.

2.2 Distributions and Random Variables

All distributions under consideration are finite. For a distribution $\mathcal{X}$, let $x \stackrel{\mathcal{R}}{\leftarrow} \mathcal{X}$ stand for the process of sampling x according to $\mathcal{X}$. Similarly, for a set $\mathcal{X}$, $x \stackrel{\mathcal{R}}{\leftarrow} \mathcal{X}$ stands for the process of sampling x uniformly from $\mathcal{X}$. For two distributions $\mathcal{X}, \mathcal{Y}$, let $\text{SD}(\mathcal{X}, \mathcal{Y})$ denote their *statistical distance*. Throughout, we naturally associate random variables with their marginal distributions. Given two interactive algorithms A and B , we let (A, B) denote their outputs in a random interaction between them. A *coupling* between distributions $\mathcal{X}$ and $\mathcal{Y}$ is a pair of (jointly distributed) random variables (X, Y) such that the marginal distribution of X is $\mathcal{X}$, and that of Y is $\mathcal{Y}$. It is well known that $\text{SD}(\mathcal{X}, \mathcal{Y}) \leq \Pr[X \neq Y]$ for any coupling (X, Y) of $(\mathcal{X}, \mathcal{Y})$. Let Ber_ε denote the Bernoulli experiment that returns 1 with probability ε and 0 otherwise.

2.3 Games

We naturally describe different security notions using two-party games. Since we also care about expected running-time algorithms, it is more natural to define game players as (uniform) algorithms. Therefore, we have to use asymptotic security notions rather than concrete security (i.e., fixed-size circuits).¹⁴

Definition 2.1 (Games). *A game G is defined by an algorithm Ch , the challenger. Algorithm P breaks the ε -security of G , if $\Pr[(P, \text{Ch})(1^\kappa) = 1] \geq \varepsilon(\kappa)$ for infinitely many κ 's. Algorithm P breaks the (ε, t, d) -security of G , if the following holds:*

Success probability: P breaks the ε -security of G .

Running time: If $d = 0$, then $P(1^\kappa)$ runs in time at most $t(\kappa)$. If $d \geq 1$, then $\mathbb{E}[T_\kappa^d] \leq t(\kappa)$, for T_κ denoting P 's running time in $(P, \text{Ch})(1^\kappa)$.

The game is (ε, t, d) -secure, if no algorithm breaks its (ε, t, d) -security. When $d = 0$, we typically omit d from the notation,

¹⁴Nevertheless, all technical results presented apply to circuits.

The discrete-logarithm game. We will pay special attention to reductions to the discrete-log game, which is naturally defined as follows.

Definition 2.2 (The discrete-logarithm game). *The discrete-log game over an ensemble of $s(\kappa)$ -size cyclic groups $\mathcal{G} = \{\mathcal{G}_\kappa\}$, each generated by $g_\kappa \in \mathcal{G}_\kappa$, is defined by the challenger that on input 1^κ :*

1. *Sends $y = g_\kappa^x$ to P , for $x \xleftarrow{R} [s(\kappa)]$.*
2. *Accepts if P sends back x .*

2.3.1 Oracle-Aided Games

We also consider oracle-aided games, defined with respect to a randomly sampled function (i.e., the random oracle).

Definition 2.3 (Oracle games). *Oracle game $G^{\mathcal{H}}$ is defined by an oracle-aided algorithm Ch , the challenger, and a function family ensemble $\mathcal{H} = \{\mathcal{H}_\kappa\}$, the oracle. Algorithm P breaks the ε -security of G , if $\Pr_{h \leftarrow \mathcal{H}_\kappa}[(P^h, \text{Ch}^h)(1^\kappa) = 1] \geq \varepsilon(\kappa)$ for infinitely many κ 's. Algorithm P breaks the (ε, t, d) -security of G , if the following holds:*

Success probability: P breaks the ε -security of G .

Running time: *If $d = 0$, then $P(1^\kappa)$ runs in time at most $t(\kappa)$ (for any $h \in \mathcal{H}_\kappa$). If $d \geq 1$, then $\mathbb{E}[R_\kappa^d] \leq t(\kappa)$, for R_κ denoting P 's running time in $(P^H, \text{Ch}^H)(1^\kappa)$ for $H \xleftarrow{R} \mathcal{H}_\kappa$.*

2.4 Sigma Protocols

Definition 2.4 (Sigma protocols). *A Sigma protocol for relation $\mathcal{R}$ is a three-message, public-coin, perfect honest-verifier zero-knowledge protocol for $\mathcal{R}$.*

Note that we do not require special soundness. We sometimes refer to the verifier's message as the *challenge query*. The above stipulates that the honest-verifier zero-knowledge property holds perfectly; yet, our results readily extend to statistical zero knowledge. We denote the challenge domain from which the prover samples its challenges on input of size κ by $\mathcal{E}_\kappa$, and address $\mathcal{E} = \{\mathcal{E}_\kappa\}$ as its *challenge space*.

We also use the following notation.

Notation 2.5 (Sigma protocols). *We view the prover P of a Sigma protocol as a pair of algorithms (P_1, P_2) .*

- $P_1(x, w; r)$: *Output P 's first message a induced by input x , witness w and randomness r .*
- $P_2(x, w, e; r)$: *Output P 's second message induced by the input x , witness w , randomness r and challenge message e .*

The following game captures the security of a Sigma protocol.

Definition 2.6 (The Sigma protocol game). *Given a Sigma protocol $\Sigma = (P, V)$ and relation $\mathcal{R}$, the interactive game $G_{\Sigma, \mathcal{R}}$ is defined by the challenger that on input 1^κ :*

1. *Sends $x \xleftarrow{R} \mathcal{L}(\mathcal{R}) \cap \{0, 1\}^\kappa$ to P .*
2. *Continues as $V(x)$.*

Identification schemes. The *key-only security* of an identification scheme is a weaker notion of security where the attacker is only given the verification key and has to impersonate the prover. It is immediately clear that the key-only security of a Sigma protocol-based identification schemes with respect to Sigma protocol $\Sigma = (\mathsf{P}, \mathsf{V})$ and relation $\mathcal{R}$ is captured by the security of the game $\mathsf{G}_{\Sigma, \mathcal{R}}$. Hence, our lower bounds on reductions that prove the security of Sigma-protocol games apply to Sigma-protocol-based identification schemes, and in particular to the Schnorr identification scheme.

2.4.1 Non-interactive Sigma protocols

By non-interactive Sigma protocols, we refer to the Fiat-Shamir variant of the Sigma protocol: the challenge message e is sampled by the prover using the “hash function”.

Definition 2.7 (Non-interactive Sigma protocols). *Given a sigma protocol $\Sigma = (\mathsf{P}, \mathsf{V})$, relation $\mathcal{R}$ and a function h , the non-interactive protocol $\mathsf{G}_{\Sigma, \mathcal{R}}^h = (\mathsf{P}^h, \mathsf{V}^h)$ is defined as follows:*

P^h : On input (x, w) :

1. Let $(a, r) \xleftarrow{\mathcal{R}} \mathsf{P}^1(x, w)$.
2. Let $e \leftarrow h(x, a)$.
3. Let $z \leftarrow \mathsf{P}^2(x, w, r, e)$.
4. Output (a, z) .

V^h : On input (x, a, z) :

1. Let $e \leftarrow h(x, a)$.
2. Output $\mathsf{V}(x, a, e, z)$.

Definition 2.8 (The non-interactive Sigma protocol game). *Let $\Sigma = (\mathsf{P}, \mathsf{V})$ be Sigma protocol with challenge space $\mathcal{E}$ and let $\mathcal{R}$ be a relation. The Fiat-Shamir game $\mathsf{G}_{\Sigma, \mathcal{R}}^{\text{FS}}$ is the oracle-game define by the function family ensemble $\{\mathcal{H}_\kappa: \{0, 1\}^* \mapsto \mathcal{E}_\kappa\}$ and by challenger that on input 1^κ and oracle access to h :*

1. Send $x \xleftarrow{\mathcal{R}} \mathcal{L}(\mathcal{R}) \cap \{0, 1\}^\kappa$ to P .
2. Continues as $\mathsf{V}^h(x)$.

Signature schemes. The key-only security of the Sigma protocol based signature schemes with respect to Sigma protocol $\Sigma = (\mathsf{P}, \mathsf{V})$ and relation $\mathcal{R}$ is simply the security of the game $\mathsf{G}_{(\mathsf{P}, \mathsf{V}), \mathcal{R}}^{\text{FS}}$, were on response (m, σ) send by P , V^h returns $\mathsf{V}^{h_m}(\sigma)$ for $h_m(z) := h(m, z)$. Hence, our bounds on the security of non-interactive Sigma protocol games also hold for Sigma protocol-based signature schemes, and in particular, for the *Schnorr signature scheme*.

2.5 Security Reductions

Definition 2.9 (Black-box security reductions). *Algorithm R is a black-box reduction from the (ε_2, t_2) -security of game G_2 to the (ε_1, t_1, d) -security of game G_1 if for any algorithm A that ε_2 -breaks the security of G_2 , R^A (ε_1, t_1, d) -breaks the security of G_1 , where we charge $t_2(\kappa)$ steps for each call to $A(1^\kappa)$. In addition, on input 1^κ , R only invokes A on input 1^κ and can*

1. Query A on arbitrary values.
2. Rewind A : instruct A to set its state to that before the last call.

Remark 2.10. *Note that we only consider reductions from the (ε_2, t_2) -security of a game G_2 , rather than the more general (ε_2, t_2, d) -security (in contrast to game G_1 , where we leave d as a parameter of the reduction). In other words, we focus on reductions that prove security against worst-case adversaries, but the reduction is allowed to transform the worst-case attacker for G_2 into a “higher moment” attacker for G_1 . This focus is justified because security reductions typically argue that worst-case attackers can break the game in question. Additionally, it allows us to focus on a narrower set of reductions, thus simplifying the text and alleviating notation.*

Reductions from oracle games. The notion of black-box reductions naturally extends to the case where the games are defined with respect to some function family.

Definition 2.11 (Black-box reductions from oracle-games). *Algorithm R reduces the $(\varepsilon_2, t_2, q_2)$ -security of an oracle-game $G_2^{\mathcal{H}}$ to the (ε_1, t_1, d) -security of a game G_1 if for any q_2 -query algorithm A that ε_2 -breaks the security of $G_2^{\mathcal{H}}$, R^A (ε_1, t_1, d) -breaks the security of G_1 , where we charge $t_2(\kappa)$ steps for each call to A . In addition, on input 1^κ , R only invokes A on input 1^κ and can*

1. Query A on arbitrary values.
2. Rewind A : instruct A to set its state to that before the last call.
3. Observe and answer A ’s oracle calls.

The reduction is programming if it answers the oracle calls arbitrarily. It is weakly programming if it answers the oracle’s calls uniformly: it answers a query x with $h(x)$ for $h \xleftarrow{R} \mathcal{H}_\kappa$, or with one of the answers it gave before on x . (In both cases, the reduction may provide different answers to distinct queries on the same value x .)

Note that, in this case, we also do not consider the full generality of parameters, but instead focus on the most relevant choice.

3 Emulating Uniform-Challenge Reductions

In this section, we present our main conceptual contribution, showing how to emulate uniform-challenge reductions that prove the security of Sigma protocols. In Section 3.1, we consider interactive protocols, and in Section 3.2, we consider non-interactive protocols.

3.1 Reductions from Interactive Sigma Protocols

We begin by formally defining uniform-challenge reductions from (interactive) Sigma protocols.

Definition 3.1 (Uniform-challenge reductions from Sigma protocols). *Let Σ be a Sigma protocol and let R be a black-box reduction from the security of G_Σ to some other game. Algorithm R is uniform-challenge, if the queries it sends to an adversary are uniform sampled over the challenge space of Σ . Specifically, on input 1^κ , each query is (independently) sampled as follows:*

1. Sample fresh random coins (on a special tape).
2. Use these coins to sample $e \in \mathcal{E}_\kappa$, for $\mathcal{E}$ being the challenge space of Σ .

For any Sigma protocol, we present an inefficient adversary that breaks its security, and yet a uniform-challenge reduction cannot exploit access to this adversary by too much. To make the text more readable, we construct an adversary that does not require rewinding: it accepts two types of queries: input-query such that on value x the adversary is instructed to set its input to x , and challenge-query, such that on input e the adversary is instructed to answer the challenge e . For the same reason, we also instruct the adversary to output the full transcript in response to a successful challenge-query.

Algorithm 3.2. [A: Adversary for (interactive) Sigma protocols]

Parameters: Sigma protocol Σ , relation $\mathcal{R}$, $\kappa \in \mathbb{N}$ and $\varepsilon \in [0, 1]$.

Let $(P = (P_1, P_2), \cdot)$ be the prover's in Σ , let $\mathcal{D}$ denote P 's randomness domain^a and let $\mathcal{E}$ denote the and verifier's challenge space.

On input-query x :

1. If $x \notin \mathcal{L}(\mathcal{R}) \cap \{0, 1\}^\kappa$, return $\perp$.
2. Find (using brute force) w such that $(x, w) \in \mathcal{R}$.
3. Sample $m_G: \mathcal{E}_\kappa \mapsto \{0, 1\}$, by (independently) sampling, for each e , $m_G(e) \stackrel{R}{\leftarrow} \text{Ber}_\varepsilon$.
4. Sample $r \stackrel{R}{\leftarrow} \mathcal{D}_\kappa$ and output $a \leftarrow P_1(x, w; r)$.

On challenge-query e : If $m_G(e) = 1$, output (x, a, e, z) for $z \leftarrow P_2(x, w, e; r)$. Else, output $\perp$.

^aOn input $x \in \{0, 1\}^\kappa$, P_1 samples its random coins uniformly from $\mathcal{D}_\kappa$.

Note that for each $x \in \mathcal{L}(\mathcal{R})$, **A** makes the verifier accept with probability ε .

Theorem 3.3 (Emulating uniform-challenge reductions). *Let Σ be a Sigma protocol for a relation $\mathcal{R}$ and let R be a uniform-challenge reduction from the security of G_Σ to a game G . Then there exists an algorithm $\hat{R}$ such that, for any $\kappa, v \in \mathbb{N}$, there exists a coupling between the coins of $\hat{R} = \hat{R}_{\kappa, v}$, and that of $R = R_\kappa$ and $\mathbf{A} = \mathbf{A}_{\Sigma, \mathcal{R}, \kappa, v}$, such that the following holds for any fixing of G 's coins: let **Double** be the event that in the execution of $(R^{\mathbf{A}}, G)(1^\kappa)$ algorithm R asks two challenge-queries resulting in (x, a, e^0, z^0) and (x, a, e^1, z^1) with $e^0 \neq e^1$.*

1. The execution of $(\hat{R}, G)(1^\kappa)$ and $(R^{\mathbf{A}}, G)(1^\kappa)$ are identical until **Double** occurs.

2. The running time of $\widehat{R}$'s in $(\widehat{R}, G)(1^\kappa)$ is at most that of R in $(R^A, G)(1^\kappa)$, plus (1) $(t_R + t_{\text{Sim}})$ additional steps per each input-query R makes to A before Double, for t_R being the time to compute membership in $\mathcal{L}(\mathcal{R}) \cap \{0, 1\}^\kappa$ and t_{Sim} being the running time to the guaranteed zero-knowledge simulator of Σ on inputs in $\{0, 1\}^\kappa$. (2) $O(\log(|\mathcal{E}_\kappa|))$ steps per each challenge-query, for $\mathcal{E}$ being the challenge-space of Σ .

That is, $\widehat{R}$ runs slightly more than R^A (without charging for the running time of A), and the probability that R^A outperforms $\widehat{R}$ is bounded by the probability that R^A makes two “hitting” challenge queries.

Proof. Fix $\Sigma, \mathcal{R}, G, R$, and let Sim be the zero-knowledge simulator guaranteed for Σ . The emulator $\widehat{R}$ is defined as follows:

Algorithm 3.4. [Emulator $\widehat{R}$]

Parameters: $\kappa, \varepsilon \in \mathbb{N}$.

Operation: Emulate a random execution of R^A as follows:

- On input-query x :
 1. If $x \notin \mathcal{L}(\mathcal{R}) \cap \{0, 1\}^\kappa$, return $\perp$.
 2. Let $(a, e, z) \xleftarrow{R} \text{Sim}(x)$.
 3. Sample $m_G: \mathcal{E}_\kappa \mapsto \{0, 1\}$ as done by A .^a If $\mathcal{G} = \{e' \in \mathcal{E}_\kappa: m_G(e') = 1\} \neq \emptyset$ and $e \notin \mathcal{G}$:
 - (a) For $e' \xleftarrow{R} \mathcal{G}$, set $m_G(e') \leftarrow 0$.
 - (b) Set $m_G(e) \leftarrow 1$.
 4. Set $e_{\text{remap}} \leftarrow \perp$.
 5. Set A 's answer to a .
- On a challenge-query (following a non- $\perp$ input-query):
 1. Sample $e' \xleftarrow{R} \mathcal{E}_\kappa$.

If $m_G(e') = 0$, set R 's challenge to e' and A 's answer to $\perp$.

Else,

Abort if $e_{\text{remap}} \notin \{\perp, e'\}$. (Double occurs.)

Else,

 - (a) Set $e_{\text{remap}} \leftarrow e'$.
 - (b) Set R 's challenge to e and A 's answer to (x, a, e, z) .

^aAs stated, this step makes $\widehat{R}$ inefficient. To adjust its code so that its efficiency matched the theorem statement, the function m_G is sampled on the fly.

We argue that $\widehat{R}$ honestly emulates A until Double occurs. Assume for ease of notation that $R(1^\kappa)$ only makes a single input-query $x \in \mathcal{L}(\mathcal{R}) \cap \{0, 1\}^\kappa$ to A , and that R is deterministic apart from the

coins it uses for sampling the challenge-queries. The coupling between the coins of $\widehat{R}$, and the coins of R and A , is rather straightforward:

$\widehat{R}$'s coins: Sample them uniformly at random.

A 's coin's: The function m_G is set to the value sampled by $\widehat{R}$.

R 's coins: The challenge-queries are set according to the queries output by the execution of $\widehat{R}$. After $\widehat{R}$ aborts, if at all, the next query is set to the value e' that caused the abort, and the following queries are sampled uniformly at random.

Fix the coins of G . Since for $(\cdot, e, \cdot) \stackrel{R}{\leftarrow} \text{Sim}(x)$ it holds that e is uniform in $\mathcal{E}_\kappa$, the function m_G sampled by $\widehat{R}$ is distributed like in A . It is also not hard to see that the challenge-queries output by $\widehat{R}$ are uniform and identically distributed in $\mathcal{E}_\kappa$: the value of the challenge-query is altered at most once, and if done, it is from a uniformly sampled element in $\mathcal{E}_\kappa \cap m_G^{-1}(1)$ to a uniformly sampled element in $\mathcal{E}_\kappa \cap m_G^{-1}(1)$. Hence, the above coupling is perfect until $\widehat{R}$ aborts. We conclude the proof by noticing that $\widehat{R}$ aborts in the above coupling iff **Double** occurs in the execution of (R^A, G) defined by the above coupling on the same challenge-query. $\square$

3.2 Reductions from *Non-Interactive Sigma Protocols*

For any Sigma protocol, we construct an inefficient adversary that breaks the corresponding non-interactive game, yet any weakly-programmable reduction cannot benefit from access to this adversary by “too much”. As in the interactive case, the adversary is defined so that rewinding is not required.

Notation 3.5. For $\varepsilon \in [0, 1]$ and $b \in \mathbb{N}$, let $\delta(\varepsilon, b) \in [0, 1]$ be such that $1 - (1 - \delta(\varepsilon, b))^b = \varepsilon$.

Algorithm 3.6. [A : adversary for non-interactive Sigma protocols]

Parameters: Sigma protocol Σ , relation $\mathcal{R}$, $\kappa \in \mathbb{N}$, $\varepsilon \in [0, 1]$ and $b \in \mathbb{N}$.

Let $(P = (P_1, P_2), \cdot)$ be the prover's in Σ , let $\mathcal{D}$ denote P 's randomness domain, and let $\mathcal{E}$ denote the verifier's challenge space.

Oracle: $h: \{0, 1\}^* \mapsto \mathcal{E}_\kappa$.

On input-query x : 1. If $x \notin \mathcal{L}(\mathcal{R}) \cap \{0, 1\}^\kappa$, return $\perp$.

2. Find (using brute force) w such that $(x, w) \in \mathcal{R}$.

3. Sample $m_r: \{0, 1\}^* \mapsto \mathcal{D}_\kappa$ uniformly.^a

4. Sample $m_G: \{0, 1\}^* \mapsto \{0, 1\}$, by (independently) sampling, for each u , $m_G(u) \stackrel{R}{\leftarrow} \text{Ber}_{\delta(\varepsilon, b)}$.

On challenge-query: (following input-query x that was not answered by $\perp$):

1. For $i = 1$ to b :

(a) Let $r \leftarrow m_r(e_1, \dots, e_{i-1})$.

(b) Let $a_i \leftarrow P_1(x; r)$.

- (c) Let $e_i \leftarrow h(x, a_i)$.
- (d) If $m_G(a_i, e_i) = 1$, output (x, a_i, e_i, z_i) and halt, for $z_i \leftarrow P_2(x, w, e_i; r)$.

2. Output $\perp$.

^a m_r , and also the following function m_G , is only invoked over a *finite* domain. We define it over $\{0, 1\}^*$ to avoid over-complicated notation.

Note that for each $x \in \mathcal{L}(\mathcal{R})$, $\mathbf{A}$ makes the verifier accept with probability ε . Also note that changing the answer of h on a_i changes the value of $a_{i+1}, \dots, a_b$ to random values. Doing that makes the security reduction harder; to extract a witness from the prover, it must answer two different challenges for the same committing message a_i , which requires modifying the oracle answer on a_i . As a result, the above action will change the future committing messages used by the prover, forcing the reduction to focus on exploiting a single committing message (per query).

Theorem 3.7 (Emulating weakly-programmable reductions). *Let Σ be a Sigma protocol for a relation $\mathcal{R}$, let R be a weakly programming reduction from the security of $G_{\Sigma, \mathcal{R}}^{\text{FS}}$ to a game G . Then, there exists an algorithm $\widehat{R}$ such that for any $\kappa, b \in \mathbb{N}$ and $\varepsilon \in [0, 1]$, there exists a coupling between the coins of $\widehat{R} = \widehat{R}_{\kappa, \varepsilon, b}$, and that of $R = R_\kappa$ and $\mathbf{A} = \mathbf{A}_{\Sigma, \mathcal{R}, \kappa, \varepsilon, b}$, such that the following holds for any fixing of G 's coins. Let **Double** be the event that in the execution of $(R^{\mathbf{A}}, G)$, algorithm R asks two challenge-queries resulting in $(\cdot, a, e^0, z^0)$ and $(\cdot, a, e^1, z^1)$ with $e^0 \neq e^1$. Then, there exist a coupling between $(R^{\mathbf{A}}, G)$ and $(\widehat{R}, G)$ such that:*

1. *The execution of $(\widehat{R}, G)(1^\kappa)$ and $(R^{\mathbf{A}}, G)(1^\kappa)$ are identical until **Double** occurs.*
2. *The running time of $\widehat{R}$'s in the coupling is at most that of $R^{\mathbf{A}}$, excluding $\mathbf{A}$'s running time, plus (1) $(t_{\mathcal{R}} + t_{\text{sim}})$ additional steps per each input-query R makes to $\mathbf{A}$ before **Double**, for $t_{\mathcal{R}}$ being the time to compute membership in $\mathcal{L}(\mathcal{R}) \cap \{0, 1\}^\kappa$ and t_{sim} being the running time to the guaranteed zero-knowledge simulator of Σ on inputs in $\{0, 1\}^\kappa$. (2) $O(\log(|\mathcal{E}_\kappa|))$ steps per each challenge-query, for $\mathcal{E}$ being the challenge-space of Σ .*

Note that while the definition of **Double** above is syntactically similar to that in Theorem 3.3, since $\mathbf{A}$ samples which a to answer from b distinct queries, the probability of **Double** above is essentially $1/b$ smaller than in Theorem 3.3 (see Theorem 4.10).

Proof of Theorem 3.7. Fix $\Sigma, \mathcal{R}, G, R, b$, let $\mathcal{H}_\kappa: \{0, 1\}^* \mapsto \mathcal{E}_\kappa$ be the all-functions family, and let Sim be the zero-knowledge simulator guaranteed for Σ . The efficient algorithm $\widehat{R}$ is defined as follows.

Algorithm 3.8. [Emulator $\widehat{R}$]

Parameters: $\kappa, v, b \in \mathbb{N}$.

Operation: Emulate a random execution of $R^{\mathbf{A}}$ as follows:

- On input-query x :
 1. If $x \notin \mathcal{L}(\mathcal{R}) \cap \{0, 1\}^\kappa$, return $\perp$.
 2. Sample $m_a, m_{e, z}, m_R: \{0, 1\}^* \mapsto \{0, 1\}^*$ as follows:^a for each $u \in \{0, 1\}^*$:

- (a) Sample $(a, e, z) \xleftarrow{R} \text{Sim}(x)$.
 - (b) Set $m_a(u) \leftarrow a$.
 - (c) If $m_{e,z}(a) = \perp$, set $m_{e,z}(a) = (e, z)$.
 - (d) Set $m_{\text{remap}}(a) \leftarrow \perp$.
3. Sample $m_G: \{0, 1\}^* \mapsto \{0, 1\}$ as in **A**.
4. For each $a \in \{0, 1\}^*$:
- (a) Let $(e, \cdot) \leftarrow m_{e,z}(a)$.
 - (b) If $\mathcal{G} \leftarrow \{e' \in \mathcal{E}_\kappa: m_G(a_i, e') = 1\} \neq \emptyset$ and $e \notin \mathcal{G}$:
 - i. For $e' \xleftarrow{R} \mathcal{G}$, set $m_G(a_i, e') \leftarrow 0$.
 - ii. Set $m_G(a_i, e) \leftarrow 1$.
- On challenge-query (following input-query x that was not answered by $\perp$):
- 1. For $i = 1$ to b :
 - (a) Let $a_i \leftarrow m_a(e'_1, \dots, e'_{i-1})$.
 - (b) If $m_G(a_i, h(x, a_i)) = 0$, continue to $i + 1$. Else,
 - (c) Let $(e_i, z_i) \leftarrow m_{e,z}(a_i)$.
 - (d) If $h(x, a_i)$ is resampled:
 - i. Abort if $m_{\text{remap}}(a) \notin \{\perp, h(x, a_i)\}$.
 - ii. Set $m_{\text{remap}}(a_i) \leftarrow h(x, a_i)$.
 - iii. Set $h(x, a_i) \leftarrow e_i$.
 - (e) Return (x, i, a_i, e_i, z_i) .
 - 2. Return $\perp$.

^aSimilarly to the interactive case, it is easy to make the emulator efficient by sampling the functions on the fly.

That is, similarly to the proof of Theorem 3.3, the emulator uses the zero-knowledge simulator for the ability to answer a single query per a . Assume for ease of notation that $R(1^\kappa)$ only makes a single input-query $x \in \mathcal{L}(\mathcal{R}) \cap \{0, 1\}^\kappa$ to **A**, and that R is deterministic apart from the coins it uses for resampling the oracle. The coupling between the coins of $\widehat{R}$, and the coins of R and **A**, is rather straightforward:

$\widehat{R}$'s coins: Sampled uniformly at random.

A's coin's: The function m_G sampled by **A** is set to value sampled by $\widehat{R}$, and the function m_r is set so that $P_1(x, w; m_r(u)) = m_a(u)$ and $P_2(x, w, e_i; m_r(u)) = z_i$ for $(e_i, z_i) \leftarrow m_{e,z}(m_a(u))$.¹⁵

R's coins: The outputs of h (resampled by R in a given challenge-query call) are set according to the values in $\widehat{R}$. After $\widehat{R}$, aborts, if at all, they are sampled uniformly.

To see that the coupling is correct, we make the following observations:

¹⁵Since the simulation is perfect, m_r is well defined.

1. The functions m_r and m_G sampled by $\widehat{R}$ (on an input query) are distributed exactly like in **A**.
2. The outputs of h sampled by $\widehat{R}$ are uniformly distributed. Indeed,
 - (a) Output of $h(x, a_i)$, for each a_i , is remapped at most once (in Step 1(d)iii).
 - (b) If a remapping of $h(x, a_i)$ occurs, it is from a uniformly sampled e' with $m_G(a, e') = 1$ to a uniformly sampled e' with $m_G(a, e') = 1$. Thus, the final output is uniformly distributed.
3. Until $\widehat{R}$ aborts, for each challenge-query it returns the same value as **A** does. Indeed, by the above observation about h :
 - (a) If $\widehat{R}$ returns $\perp$ then so does **A**.
 - (b) If $\widehat{R}$ returns (x, i, a_i, e_i, z_i) then **A** returns $(x, i, a_i, h(a_i), z_i)$, and by construction, $h(a_i) = e_i$.
4. $\widehat{R}$ aborts, on a given challenge-query, iff **Double** occurs in the coupled execution of R^A (on this query).

Hence, the two executions are indeed identical until **Double** occurs. $\square$

4 Bounds on Sigma Protocols

In this section, we utilize the emulations presented in Section 3 along with bounds we provide on *double-hitting games* to establish bounds on uniform-query reductions from Sigma protocols. In Section 4.1, we prove lower bounds for uniform-challenge reductions from interactive Sigma protocol games, which readily carry over to reductions from Sigma protocol-based identification schemes (see Section 2.4). In Section 4.2, we prove lower bounds for weakly programmable reductions from non-interactive Sigma protocol games, which readily carry over to reductions from Sigma protocol-based signature schemes (see Section 2.4.1). Throughout, we use the following notation.

Notation 4.1. For an (honest-verifier zero-knowledge) Sigma protocol Σ , let $t_{\text{Sim}(\Sigma)}(\kappa)$ denote the running time of its zero-knowledge simulator on inputs in $\{0, 1\}^\kappa$. For a relation $\mathcal{R}$, let $t_{\mathcal{R}}(\kappa)$ denote the running time for computing membership in $\mathcal{L}(\mathcal{R}) \cap \{0, 1\}^\kappa$. Let $t_{E(\Sigma)}(\kappa)$ denote the $O(\log(|\mathcal{E}_\kappa|))$ steps it takes $\widehat{R}$ from Theorem 3.3 to emulate a challenge-queries with respect to sigma protocol Σ of challenge space $\mathcal{E}$.

For ease of notation, we assume that for ε_Σ , the security the reduction is aiming for, and any $\kappa \in \mathbb{N}$, it holds that $\varepsilon_\Sigma(\kappa) \cdot |\mathcal{E}_\kappa| \in \mathbb{N}$, for $\mathcal{E}$ being the challenge space of Σ .

4.1 Interactive Sigma Protocols

In Section 4.1.1, we introduce the *double-hitting game* that captures the probability for uniform-challenge reductions to cause the event **Double** (defined in Section 3), and bound the value of this game for different types of adversaries. In Section 4.1.2, we use this bound together with the emulation theorem proved in Section 3.1, to get our lower bound on uniform-query reductions from the security of (interactive) Sigma protocols.

4.1.1 The Double Hitting Game

The double-hitting game is defined as follows:

Game 4.2. [Double hitting]

Parameters: $\varepsilon \in [0, 1]$.

Parties: Challenger C and player P.

The game:

1. P (adaptively) queries C, and gets $c \xleftarrow{R} \text{Ber}_\varepsilon$.
2. P **wins** if the value 1 is returned for at least *two* queries.

Namely, the player has to obtain two successful trials. It is clear that a player of worst-case query complexity q wins the game with probability at most $q^2 \varepsilon^2$. The following theorem, proved in Section 5, generalizes this bound to players with bounded high-moment query complexity.

Notation 4.3 (Double hitting). *Let G be according to Game 4.2. Let $\text{DH}(\varepsilon, q) = \text{DH}(0, \varepsilon, q)$ denote the bound on the winning probability of a q -query algorithm in G_ε , where G is according to Game 4.2. For $d \in \mathbb{N}$, let $\text{DH}(d, \varepsilon, q)$ denote the bound on the winning probability of an algorithm with $E[Q^d] \leq q$ algorithm in G_ε , where Q denotes the algorithm's query complexity in a random interaction of G_ε*

Theorem 4.4 (Upper bounds on the double-hitting game). *For any $\varepsilon \in [0, 1]$ and $q \in \mathbb{R}^+$:*

Worst-case query complexity. $\text{DH}(\varepsilon, q) \leq q^2 \cdot \varepsilon^2$.

Higher-moment query complexity. *For any $d \geq 2$: $\text{DH}(d, \varepsilon, q) \leq (q^{1/d} \varepsilon)^{2-1/d}$.*

The bound for worst-case query complexity followed by a straightforward union bound. Bonding the winning probability of higher-moment query complexity algorithms is more complicated, and is proved in Section 5. We also prove that the tightness of these bounds by providing matching lower bounds, i.e., algorithms,

4.1.2 The Bounds

The key observation is that the probability the reduction R considered in Theorem 3.3 causes Double, i.e., makes two (successful, distinct) queries answered by (x, a, e, z) and (x, a, e', z') with $e \neq e'$, is at most $\text{DH}(d, \varepsilon_\Sigma, q_G)$. Hence, Theorem 3.3 easily yields the following bound.

Theorem 4.5 (Emulating uniform-challenge reductions to arbitrary games). *Let Σ be a Sigma protocol, let $\mathcal{R}$ be a relation and assume there exists a uniform-challenge reduction R from the $(t_\Sigma, \varepsilon_\Sigma)$ -security of $G_{\Sigma, \mathcal{R}}$ to the (d, t_G, ε_G) -security of a game G such that $\max\{t_{\text{Sim}(\Sigma)}(\kappa) + t_{\mathcal{R}}(\kappa), t_{\text{E}(\Sigma)}(\kappa)\} \leq t_\Sigma(\kappa)$, for any κ . Then there exists a (no-oracle) algorithm (specifically, Algorithm 3.2) that breaks the $(t_G, \varepsilon_G - \text{DH}(d, \varepsilon_\Sigma, q_G))$ -security of G , for $q_G \leftarrow t_G/t_\Sigma^d$ if $d \geq 1$, and $q_G \leftarrow t_G/t_\Sigma$ otherwise.*

Namely, the no-oracle attacker on G only loses $\text{DH}(d, \varepsilon_\Sigma, q_G)$ compared to the reduction (that has oracle access to an adversary for $G_{\Sigma, \mathcal{R}}$).

Proof of Theorem 4.5. The proof follows by the no-oracle emulation guaranteed by Theorem 3.3 by bounding the probability of the event **Double** considered in the theorem to winning in the Game 4.2.

Let $A = A_{\Sigma, \mathcal{R}, \kappa, \varepsilon_\Sigma}$ be according to Algorithm 3.2. By Theorem 3.3, to prove the theorem it suffices to show that in a random execution of $(R^A(1^\kappa), G_{\Sigma, \mathcal{R}}(1^\kappa))$, the probability of **Double**, i.e., two of A 's answers are (x, a, e, z) and (x, a, e', z') for some (x, a, e, z, e', z') with $e \neq e'$, is at most $\text{DH}(d, \varepsilon_\Sigma(\kappa), q_G(\kappa))$. Consider the following player P for Game 4.2:

Algorithm 4.6. $[P]$ Emulates a random execution of $(R^A(1^\kappa), G_{\Sigma, \mathcal{R}}(1^\kappa))$, where on each challenge-query R makes:

1. Query C to get output c .
2. If $c = 1$, set $m_G(e) = 1$.
Else, set $m_G(e) = 1$.

By construction, the emulation of $(R^A(1^\kappa), G_{\Sigma, \mathcal{R}}(1^\kappa))$ done by P is random execution. It is also clear that the probability of **Double** in the emulated execution is at most the probability that P wins the game. Hence, since the query complexity of R , and thus of P , is at most q_G , we conclude that $\Pr[\text{Double}] \leq \text{DH}(d, \varepsilon_\Sigma(\kappa), q_G(\kappa))$. $\square$

Combining Theorems 4.4 and 4.5 yields the following bounds, proven below, that are applicable to the discrete log game.

Corollary 4.7 (Bounds on uniform-challenge reductions). *Let $G_{\Sigma, \mathcal{R}}, G, R, d, q_G$ be as in Theorem 4.5 and let $s: \mathbb{N} \mapsto \mathbb{N}$ and $c > 0$. Then:*¹⁶

Case $d = 0$. *If G is $(d, t_G, c \cdot t_G^2/s)$ -secure and $\varepsilon_G \geq 2ct_G^2/s$, then $\varepsilon_\Sigma^2 \geq ct_\Sigma^2/s$.*

Case $d \geq 2$. *If G is $(d, t_G, c \cdot t_G/s)$ -secure and $\varepsilon_G \geq 2ct_G/s$, then $\varepsilon_\Sigma^{2-1/d} \geq c \cdot q_G^{1-2/d+1/d^2} t_\Sigma^d/s$.*

Proof. By the assumption on the security of G and Theorem 4.5, we derive that $\text{DH}(d, \varepsilon_\Sigma, q_G) \geq ct_G/s$, proving the first part of each case. The specific bounds, per d , readily follow from the bounds in Theorem 4.4. $\square$

Namely, focusing for concreteness on $d = 0$, if $\varepsilon_G \geq c \cdot t_G^2/s$, then unless the reduction proves that G is *not* $(t_G, c \cdot t_G^2/s)$ -secure, it holds that $\varepsilon_\Sigma^2 \geq ct_\Sigma^2/s$. We argue that for games in which one cannot rule out a two-factor improvement in the known upper bounds, i.e., algorithms, focusing on reductions with ε_G that is twice the known upper bound is sufficient; reductions with smaller ε_G are only meaningful if we believe such an improvement does not exist. Therefore, for reductions that do not improve the known upper bound and do not rely on the fact that the known upper bounds are tight up to a two-factor, our bound is applicable.

Bounds on Schnorr identification schemes. As discussed in Section 2.4, the above bounds hold for reductions from the security of Schnorr identification schemes. Consider such reductions to the security of the discrete log game (see Definition 2.2). Note that for such reductions, it is always the case that $\max\{t_{\text{Sim}(\Sigma)}(\kappa) + t_{\mathcal{R}}(\kappa), t_{\text{E}(\Sigma)}(\kappa)\} \leq t_\Sigma(\kappa)$ (otherwise the reduction only addresses

¹⁶In the following, we ignore the case $d = 1$, since the hardness assumption in this case is not useful. See Section 1.1.1.

adversary of some small (fixed) polynomial running time). For $d = 0$, i.e., worst-case running time, the upper-bound in Corollary 4.7 matches the lower-bound of [PS00]. For $d > 1$, the upper-bound in Corollary 4.7 matches the lower-bound of [RS24].

4.2 Non-Interactive Sigma Protocols

In Section 4.2.1, we introduce the *coins and cows game*¹⁷ that captures the probability for weakly programmable reductions to cause the event Double (defined in Section 3), and bound the value of this game for different types of adversaries. In Section 4.2.2, we utilize this bound in conjunction with the emulation theorem proved in Section 3.2 to derive our lower bound on weakly programmable reduction for the security of non-interactive Sigma protocols.

4.2.1 The Coins and Cows Game

The coins and cows game is defined as follows:

Game 4.8. [Coins and cows]

Parameters: $\varepsilon \in [0, 1]$ and $b, s \in \mathbb{N}$.

Parties: Challenger C and player P.

The game:

1. C: sets $w \leftarrow 0$.
2. On each query made by P, C acts as follows:
 - (a) Set $c' \leftarrow 0$ and sample $c \stackrel{R}{\leftarrow} \text{Ber}_\varepsilon$.
 - (b) If $c = 1$:
 - i. If $w \geq 1$: sample $c' \stackrel{R}{\leftarrow} \text{Ber}_{\varepsilon'}$ for $\varepsilon' \leftarrow \frac{1}{b} + \frac{w}{s}$.^a
 - ii. Set $w \leftarrow w + 1$.
 - (c) Send (c, c') to P.
3. P wins if at some point it holds that $c' = 1$.

^aWe use the convention that Ber_δ for $\delta > 1$ equals Ber_1 , i.e., the constant one distribution.

Namely, effectively, the player has to hit twice, but after the first hit the success probability decays to ε' instead of ε , where ε' is ε/b plus a correction that depends on the number of hits so far and the parameter s . The following theorem, proved in Section 5, bounds the value of the above game for worst-case running-time and for higher-moment running time players.

Notation 4.9 (Coins and cows). Let $\text{RDH}(\varepsilon, q, b) = \text{RDH}(0, \varepsilon, b, s, q)$ denote the bound on the winning probability of a q -query algorithm in $\mathcal{G}_{\varepsilon, b, s}$. For $d \in \mathbb{N}$, let $\text{RDH}(d, \varepsilon, b, s, q)$ denote the

¹⁷The name is a playful homage to the classic pen-and-paper game “bulls and cows” (also known as “bulls and pigs”) that the introduced game somewhat resembles, though still differs substantially.

bound on the winning probability of an algorithm with $E[Q^d] \leq q$ algorithm in $G_{\varepsilon,b,s}$, where Q denotes the algorithm's query complexity in a random interaction of $G_{\varepsilon,b,s}$

Theorem 4.10 (Upper bounds on the coins-and-cows game¹⁸). *For $\varepsilon \in [0, 1]$, $b, s \in \mathbb{N}$ and $q \in \mathbb{R}^+$: Worst-case query complexity.*

$$\text{RDH}(\varepsilon, b, s, q) \leq \begin{cases} q^2 \cdot \varepsilon^2 \cdot (1/b + q/s) & \text{if } q\varepsilon < 1 \\ q \cdot \varepsilon \cdot (1/b + q/s) & \text{otherwise} \end{cases}.$$

Higher-moment query complexity. For $d \geq 2$:

$$\text{RDH}(d, \varepsilon, b, s, q) \leq \begin{cases} \frac{2u(1+ub/s)}{b} \cdot (q^{1/d} \cdot \varepsilon)^{2-1/d} & \text{if } q^{1/d} \cdot \varepsilon < 1 \\ \frac{2u(1+ub/s)}{b} \cdot q\varepsilon^d & \text{otherwise,} \end{cases}$$

$$\text{letting } u \leftarrow 16 \cdot (48 \cdot (d+1) \cdot \ln((s+1)/\varepsilon))^d.$$

Namely, the bounds are essentially those of the double-hitting game times $1/b$. Since the bounds on the double-hitting game are tight, it is straightforward that so are the above bounds (up to the factor of u in the $d \geq 2$ case).

4.2.2 The Bounds

The key observation of this section is that the probability the reduction R considered Theorem 3.7 causes **Double**, i.e., makes two successful distinct queries answered by (x, i, a, e, z) and (x, i, a, e', z') with $e \neq e'$, is bounded by $\text{RDH}(d, \varepsilon_\Sigma, b, s, q_G)$, for b being essentially q_Σ , and s being essentially the domain size the prover's first (i.e., commitment) message.

Let δ be according to Notation 3.5, and for a sigma protocol $\Sigma = (P, \cdot)$ and a relation $\mathcal{R}$, let $s_{\Sigma, \mathcal{R}}(\kappa) := \min_{(x,w) \in \mathcal{R}: x \in \{0,1\}^\kappa} \{2^{\text{H}_\infty(P_1(x,w))}\}$, i.e., $1/s_{\Sigma, \mathcal{R}}(\kappa)$ is the probability of the most likely first message output by P .

Theorem 4.11 (Emulating weakly programmable reductions to arbitrary games). *Let Σ be a Sigma protocol, let $\mathcal{R}$ be a relation, and assume there exists a weakly programmable reduction R from the $(t_\Sigma, \varepsilon_\Sigma, q_\Sigma)$ -security of $G_{\Sigma, \mathcal{R}}^{\text{FS}}$ to the (d, t_G, ε_G) -security of a game G such that $\max\{t_{\text{Sim}(\Sigma)}(\kappa) + t_{\mathcal{R}}(\kappa), t_{\text{E}(\Sigma)}(\kappa)\} \leq t_\Sigma(\kappa)$, for any κ .*

Then there exists a (no-oracle) algorithm (specifically, Algorithm 3.6) that breaks the $(t_G, \varepsilon_G - \text{RDH}(d, \varepsilon_\Sigma, b, s_{\Sigma, \mathcal{R}}, q_G))$ -security of G , for $q_G \leftarrow t_G/t_\Sigma^d$ if $d \geq 1$, and $q_G \leftarrow t_G/t_\Sigma$ otherwise, and $b \leftarrow \delta(\varepsilon_\Sigma, q_\Sigma)/\varepsilon_\Sigma$.

Namely, the no-oracle attacker on G does only lose $\text{RDH}(d, \varepsilon_\Sigma, q_G, q_\Sigma, s_{\Sigma, \mathcal{R}})$ compared to the reduction (that has oracle access to an adversary for $G_{\Sigma, \mathcal{R}}^{\text{FS}}$).

Proof of Theorem 4.11. The proof follows from the no-oracle emulation guaranteed by Theorem 3.7 by bounding the probability of the event **Double** considered in the theorem to winning in the Game 4.8.

Let $A = A_{\Sigma, \mathcal{R}, \kappa, \varepsilon_\Sigma, b}$ for $b \leftarrow s_{\Sigma, \mathcal{R}}(\kappa)$, be according to Algorithm 3.6. By Theorem 3.7, to prove the theorem it suffices to show that in a random execution of $(R^A(1^\kappa), G_{\Sigma, \mathcal{R}}(1^\kappa))$, the probability of **Double**, is at most $\text{RDH}(d, \varepsilon_\Sigma, b, s_{\Sigma, \mathcal{R}}, q_G)$. Consider the following player P for Game 4.8:

¹⁸The term q/s in the bound can be significantly reduced for most choices of q and ε , but we state the bound as above since, in our application, s dominates the other parameters.

Algorithm 4.12. [P] Emulates a random execution of $(R^A(1^\kappa), G_{\Sigma, \mathcal{R}}(1^\kappa))$, where on each challenge-query the R makes:

1. Let μ be the probability, over the resampled h , of the event H : A^h replies with $(\cdot, a, e, \cdot)$ for which he never answered before with $(\cdot, a, e, \cdot)$ (i.e., possibly with different x, i, z).
2. Query C to get output $(c_P, \cdot)$.
3. If $c_P = 1$, sample a coin $c_R \xleftarrow{R} \text{Ber}_{\mu/\varepsilon_\Sigma}$.
Else, set $c_R \leftarrow 0$
4. If $c_R = 1$, sample h conditioned on H .
Else, sample h conditioned on $\neg H$.

We first prove that it is always the case that $\mu \leq \varepsilon_\Sigma$, and thus algorithm P is well defined and the emulation of $(R^A(1^\kappa), G_{\Sigma, \mathcal{R}}(1^\kappa))$ done by P is uniformly distributed. Indeed, to get a non-bot answer $(\cdot, a, e, \cdot)$ you need $h(\cdot, a) \in m_G(a)$. To get a e such that $(\cdot, a, e, \cdot)$ was not replied before, it is required that h is resampled on the point a . If this is the case, the probability that $h(\cdot, a) \in m_G(a)$ is ε_Σ . We conclude the proof by showing that the probability of Double in the emulated execution of $(R^A(1^\kappa), G_{\Sigma, \mathcal{R}}(1^\kappa))$, is at most the probability that the P wins the game. We prove it by showing that this holds for each challenge-query made by R, assuming that neither Double nor “P wins the game” has occurred yet.

Let w_P denote the number of times P got the answer $c_P = 1$ until this call, and let w_R denote the number of non- $\perp$ challenge-queries R made (counting only once answers with the same value of (a, e)). By construction, it holds that $w_R \leq w_P$. We assume without loss of generality that $w_R \geq 0$, as otherwise the probability of Double is zero. Let $a_0, \dots, a_{q'}$ be the h queries ask by A in the previous challenge-query (note that, in the case A did not return $\perp$, q' might be smaller than q_Σ) and let i^* be the smallest index for which $h(a_{i^*})$ is asked to be resampled by R. We assume without loss of generality that $i^* \in [q_\Sigma]$, i.e., at least one query is resampled, as otherwise the probability of Double is zero. We make the following observations:

1. Since A returns the first query a_i with $m_G(a_i) = 1$, it is not going to return $(\cdot, i, \cdot)$ with $i < i^*$.
2. Since $h(i^*)$ is resampled, A returns $(\cdot, i^*, \cdot)$ with probability $\delta(\varepsilon_\Sigma, q_\Sigma)$.
3. Since $a_{i^*+1}, \dots, a_{q_\Sigma}$ were resampled, the probability that $m_G(a_i) = 1$ for some $i > i^*$ is bounded by $1 - (1 - \delta(\varepsilon_\Sigma, q_\Sigma))^{q_\Sigma} \leq \varepsilon_\Sigma$. If it does happen, since a_i is sampled according to $P_1(x)$, the probability that a_i appears as part of a previous answer is $w_R/s_{\Sigma, \mathcal{R}}$.

We conclude that the probability that Double occurs in this call is at most $\delta(\varepsilon_\Sigma, q_\Sigma) + \varepsilon_\Sigma \cdot w_R/s_{\Sigma, \mathcal{R}}$, and thus at most the winning probability of P in this round. $\square$

The following bounds are immediate applications of Theorems 4.10 and 4.11.

Corollary 4.13 (Bounds on weakly programmable reductions). *Let $G_{\Sigma, \mathcal{R}}, G, R, d, q_G$ be as in Theorem 4.11, let $s: \mathbb{N} \mapsto \mathbb{N}$ and $c > 0$. Then*

Case $d = 0$. If G is $(d, t_G, c \cdot t_G^2/s)$ -secure and $\varepsilon_G \geq 2c \cdot t_G^2/s$, then, letting $\widehat{q}_\Sigma \leftarrow q_\Sigma(1 + q_G q_\Sigma/s)^{-1}$:

$$\begin{cases} \varepsilon_\Sigma^2 \geq \widehat{q}_\Sigma \cdot c \cdot t_\Sigma^2/s & \text{if } q_G \varepsilon_\Sigma < 1, \\ \varepsilon_\Sigma \geq \widehat{q}_\Sigma \cdot q_G \cdot c \cdot t_\Sigma^2/s & \text{otherwise.} \end{cases}$$

Case $d \geq 2$. If G is $(d, t_G, c \cdot t_G/s)$ -secure and $\varepsilon_G \geq 2ct_G/s$, then, letting u and b be according to Theorem 4.10 and Theorem 4.11, and $\widehat{b} \leftarrow b(2u \cdot (1 + ub/s))^{-1}$:

$$\begin{cases} \varepsilon_\Sigma^{2-1/d} \geq c \cdot \widehat{b} \cdot q_G^{1-2/d+1/d^2} \cdot t_\Sigma^d/s & \text{if } q_G^{1/d} \varepsilon_\Sigma < 1, \\ \varepsilon_\Sigma^d \geq c \cdot \widehat{b} \cdot t_\Sigma^d/s & \text{otherwise.} \end{cases}$$

Namely, for weakly programmable reductions that “make sense”, see the discussion that follows Corollary 4.7, the lower bound on ε_G increases by a factor of (essentially) the adversary’s query complexity compared to the interactive case.

Bounds on Schnorr signature schemes. As discussed in Section 2.4, the above bounds hold for reductions from the security of Schnorr signature schemes. Consider such reductions to the security of the discrete log game (see Definition 2.2). For $d = 0$, the upper-bound in Corollary 4.13 matches the lower-bound of [PS00]. For $d \geq 2$, the upper-bound in Corollary 4.13 matches the lower bound of [RS24], up to a multiplicative factor of $(2u \cdot (1 + uq_\Sigma/s))^{d/(2d-1)}$ in the value of ε_Σ (recall that u is logarithmic in $1/\varepsilon$ and s , and s —the size of the group—is huge compared to u and q_Σ , so $uq_\Sigma/s \approx 0$).

5 Bounds on Hitting Games

In this section, we prove the bounds on the double-hitting games introduced in Section 4. In Section 5.2 we prove lower bounds on *multi*-hitting games (a generalization of the double-hitting game). For completeness, in Section 5.3 we provide matching lower bounds on these games. In Section 5.4, we use the bound on the multi-hitting games to prove the bounds on the coins and cows game.

5.1 Preliminaries

For $i \leq i \in \mathbb{N}$, let $(j, i] = \{j+1, \dots, i\}$ and $[j, i] = \{j, j+1, \dots, i\}$. We use the following inequality:

Proposition 5.1. *Let $d, i \in \mathbb{N}$ and $\alpha, t, k \in \mathbb{R}$ such that $i, d \geq 2$, $\alpha \in [0, 1]$, $t \in [0, (1 - \alpha) \cdot (i - 1)^d]$ and $k \in [1, d]$. Then $(t + \alpha i^d)^{k/d} - (t + \alpha(i - 1)^d)^{k/d} \geq \alpha k \cdot (i - 1)^{k-1}$.*

We prove the above inequality using the following known fact:

Fact 5.2. *Let $q \in \mathbb{N}$ and $\varepsilon \in [0, 1]$ such that $q \cdot \varepsilon \leq 1/2$. Then, $(1 - \varepsilon)^q \leq 1 - q\varepsilon + q(q - 1)\varepsilon^2/2$.*

Proof. Proof follows from standard approximations, e.g., [MU17, Lemma 4.3]. $\square$

Proof of Proposition 5.1. Define $f(x) = (t + \alpha x^d)^{k/d}$. The task is to lower bound $f(i) - f(i-1)$. As f' is strictly increasing (shown further below), by mean value theorem it holds that $f(i) - f(i-1) \geq f'(i-1)$ and the lemma reduces to proving $f'(i-1) \geq \alpha k \cdot (i-1)^{k-1}$. First, note that $f'(x) = \alpha k x^{d-1} (t + \alpha x^d)^{(k-d)/d}$ and

$$\begin{aligned} f'(i-1) &= \alpha k (i-1)^{d-1} (t + \alpha (i-1)^d)^{(k-d)/d} \\ &\geq \alpha k (i-1)^{d-1} ((1-\alpha)(i-1)^d + \alpha(i-1)^d)^{(k-d)/d} \\ &= \alpha k (i-1)^{k-1}. \end{aligned}$$

It remains to show that f' is strictly increasing. As the logarithm preserves monotonicity, it suffices to check that the derivative of $\ln(f'(x))$ is positive. Indeed,

$$\begin{aligned} (\ln(f'(x)))' &= \left(\ln(\alpha k) + (d-1) \ln(x) + \frac{k-d}{d} \cdot \ln(t + \alpha x^d) \right)' \\ &= \frac{d-1}{x} + \frac{k-d}{1} \cdot \frac{\alpha x^{d-1}}{t + \alpha x^d} \\ &= \frac{d-1}{x} \left(1 - \frac{d-k}{d-1} \cdot \frac{\alpha x^d}{t + \alpha x^d} \right) \\ &\geq \frac{d-1}{x} \left(1 - \frac{d-k}{d-1} \right) \geq 0. \end{aligned}$$

□

We will also use the following basic fact

Fact 5.3. *For any $m, d \in \mathbb{N}$, it holds that $\sum_{i=1}^m i^d \geq m^{d+1}/(d+1)$.*

Proof. $\sum_{i=1}^m i^d \geq \int_0^m x^d dx = m^{d+1}/(d+1)$. □

5.2 Multi-Hitting Games

In this section, we prove lower bounds on *multi*-hitting games, a generalization of the (single-board) double-hitting game.

Game 5.4. [The multi-hitting game $G_{n,\varepsilon}$]

Parameters: $n \in \mathbb{N}, \varepsilon \in [0, 1]$.

Parties: Challenger C and player P.

The game:

1. P (adaptively) queries C, and gets $c \xleftarrow{R} \text{Ber}_\varepsilon$.
2. P **wins** if the value 1 is returned for at least n queries.

That is, rather than requiring two hits, as in the double-hitting game, we require n hits. We prove the following upper bounds on the winning probability in the multiple-hitting games.

Notation 5.5. Let $\text{MH}_n(d, q, \varepsilon)$ denote the bound on the winning probability of any algorithm with $\mathbb{E}[Q^d] \leq q$ in $\mathbf{G}_{n, \varepsilon}$, where $\mathbf{G}$ is according to Game 5.4, and Q denotes the algorithm's query complexity in a random interaction of $\mathbf{G}_{n, \varepsilon}$.

Theorem 5.6 (Upper bounds on the multiple-hitting games). For any $\varepsilon \in [0, 1]$ and $q \in \mathbb{R}^+$.

Worst-case query compl. For any $n \in \mathbb{N}$, $\text{MH}(0, \varepsilon, q) \leq \binom{q}{n} \cdot \varepsilon^n$.

Higher-moment query compl. For any $n, d \in \mathbb{N}$, $q \in \mathbb{R}^{\geq 0}$ such that $d \geq 2$:

1. $\text{MH}_n(d, \varepsilon, q) \leq (q^{1/d} \cdot \varepsilon)^{d+(1-d)(1-1/d)^{n-1}}$.
2. If $n \geq u \leftarrow 16 \cdot (48 \cdot (d+1) \cdot \ln(n/\varepsilon))^d$ then $\text{MH}_n(d, \varepsilon, q) \leq u \cdot q \cdot (\varepsilon/n)^d$.

Note that Theorem 4.4 is a special case, for $n = 2$, of the above bound. In Section 5.3, we prove that the above bounds are tight (up to the logarithmic terms in n/ε).

Theorem 5.6 is proved via the following two propositions.

Proposition 5.7. For any $\varepsilon \in [0, 1]$, $d, n \in \mathbb{N}$ with $d \geq 2$, and $q \in \mathbb{R}^{\geq 0}$:

$$\text{MH}_n(d, \varepsilon, q) \leq (q^{1/d} \cdot \varepsilon)^{d+(1-d)(1-1/d)^{n-1}}.$$

Proposition 5.8. For any $\varepsilon \in [0, 1]$, $d, \ell \in \mathbb{N}$ with $d \geq 2$ and $2^\ell \geq 24(d+1) \ln(2^\ell/\varepsilon)$, and $q \in \mathbb{R}^{\geq 0}$:

$$\text{MH}_{2^\ell}(d, \varepsilon, q) \leq q \cdot \varepsilon^d \cdot 16 \cdot \left(\frac{24 \cdot (d+1) \cdot \ln(2^\ell/\varepsilon)}{2^\ell} \right)^d.$$

Before proving the above propositions below, we first use them to prove Theorem 5.6.

Proving Theorem 5.6.

Proof of Theorem 5.6. The proof of the worst-case running time follows by a straightforward union bound. Proposition 5.7 corresponds exactly to Item 1 of Theorem 5.6. It remains to prove Item 2 of Theorem 5.6. Let $\ell = \lfloor \log(n) \rfloor$ and apply Proposition 5.8 to deduce that (since $2^\ell \geq 24(d+1) \ln(2^\ell/\varepsilon)$):

$$\begin{aligned} \text{MH}_n(d, \varepsilon, q) &\leq \text{MH}_{2^\ell}(d, \varepsilon, q) \\ &\leq q \cdot \varepsilon^d \cdot 16 \cdot \left(\frac{24 \cdot (d+1) \cdot \ln(2^\ell/\varepsilon)}{2^\ell} \right)^d \\ &\leq q \cdot \varepsilon^d \cdot 16 \cdot \left(\frac{48 \cdot (d+1) \cdot \ln(n/\varepsilon)}{n} \right)^d \\ &\leq u \cdot q \cdot (\varepsilon/n)^d \end{aligned}$$

□

Proving Proposition 5.7.

Proof of Proposition 5.7. we start with proving the case $n = 1$, and then prove for $n > 1$ by induction.

$n = 1$. Fix a player P . We assume that P is deterministic, and the proof for the randomized player follows by a Jensen inequality. We also assume without loss of generality that P halts right after it gets a hit. The proof is by induction on m , the maximal number of queries made by P . The statement clearly holds for $m = 0$. We assume it holds for $m - 1 \geq 0$ and prove it for m . Let P' be the variant of P that aborts right after the $(m - 1)$ query (if asked). Let w and w' denote the winning probability of P and P' respectively, and let Q and Q' denote the number of queries done by P and P' respectively (all in a random execution of the game). Under this notation, we have to prove that $w \leq E[Q^d]^{1/d} \cdot \varepsilon$.

Letting $\alpha := \Pr[P \text{ makes the } m^{\text{th}} \text{ query}]$, and under the induction hypothesis, it holds that $w = w' + \alpha \cdot \varepsilon \leq E[(Q')^d]^{1/d} \cdot \varepsilon + \alpha \cdot \varepsilon$. Thus, it suffices to prove

$$\begin{aligned} E[(Q')^d]^{1/d} \cdot \varepsilon + \alpha \varepsilon &\leq E[Q^d]^{1/d} \cdot \varepsilon && \Longleftrightarrow \\ \alpha &\leq \underbrace{E[Q^d]^{1/d} - E[(Q')^d]^{1/d}}_{\delta_q} && (2) \end{aligned}$$

Let $\widehat{Q}$ denote the random variable that takes the value Q when $Q \leq m - 1$, and 0 otherwise. Let $\widehat{q} \leftarrow E[\widehat{Q}^d]$. Since $\delta_q = (\widehat{q} + \alpha m^d)^{1/d} - (\widehat{q} + \alpha(m - 1)^d)^{1/d}$, Equation (2) is equivalent to

$$(\widehat{q} + \alpha m^d)^{1/d} - (\widehat{q} + \alpha(m - 1)^d)^{1/d} \geq \alpha \quad (3)$$

Noting that $\widehat{q} \leq (1 - \alpha) \cdot (m - 1)^d$, applying Proposition 5.1 with respect to $k \leftarrow 1$ yields that $(\widehat{q} + \alpha m^d)^{1/d} - (\widehat{q} + \alpha(m - 1)^d)^{1/d} \geq \alpha$, and we are done.

$n > 1$. The proof is by induction on n , recalling that the base case $n = 1$ was proven above. Assume the bound holds for $n - 1$, and fix a player P . The following random variables are defined with respect to a random interaction of P in the game $G_{n,\varepsilon}$: let Q denote the number of queries P makes, and let $q \leftarrow E[Q^d]$. For $i \in [n]$, let H_i be the event that i^{th} hit occurred, let Q_i denote the number of queries P makes after H_{i-1} occurs and until H_i occurs (including) and let $q_i \leftarrow E[Q_i^d]$. Let $\alpha_n = d + (1 - d)(1 - 1/d)^{n-1}$. Finally, for a set $\mathcal{S} \subseteq [n]$, let $Q_{\mathcal{S}} \leftarrow \sum_{i \in \mathcal{S}} Q_i$ and $q_{\mathcal{S}} \leftarrow E[Q_{\mathcal{S}}^d]$. By the induction hypothesis,

$$\Pr[H_{n-1}] \leq (q_{[n-1]}^{1/d} \cdot \varepsilon)^{\alpha_{n-1}} \quad (4)$$

Letting $q_{n|H_{n-1}} = E[Q_n^d | H_{n-1}]$, by the case $n = 1$,

$$\Pr[H_n | H_{n-1}] \leq q_{n|H_{n-1}}^{1/d} \cdot \varepsilon \quad (5)$$

It follows that

$$\begin{aligned}
\Pr[H_n] &= \Pr[H_{n-1}] \cdot \Pr[H_n | H_{n-1}] \\
&\leq \Pr[H_{n-1}] \cdot q_{n|H_{n-1}}^{1/d} \cdot \varepsilon = \Pr[H_{n-1}] \cdot (q_n / \Pr[H_{n-1}])^{1/d} \cdot \varepsilon \\
&= \Pr[H_{n-1}]^{1-1/d} \cdot q_n^{1/d} \cdot \varepsilon \\
&\leq (q_{[n-1]}^{1/d} \cdot \varepsilon / (n-1))^{\alpha_{n-1}(1-1/d)} \cdot q_n^{1/d} \cdot \varepsilon \\
&= \underbrace{((q_{[n-1]}/q)^{1/d})^{\alpha_{n-1}(1-1/d)} \cdot (q_n/q)^{1/d}}_{\delta}^{\alpha_{n-1}(1-1/d)} \cdot (q^{1/d} \cdot \varepsilon)^{\alpha_{n-1}(1-1/d)+1}.
\end{aligned} \tag{6}$$

Notice that $\delta \leq 1$ and $\alpha_{n-1}(1-1/d) + 1 = \alpha_n$. This concludes the proof. $\square$

Proving Proposition 5.8.

Proof of Proposition 5.8. First, by Proposition 5.7, observe that for any constant $c > 1$, there exists $n_0 \in \mathbb{N}$ such that $\text{MH}_n(d, \varepsilon, q) \leq c \cdot q \cdot \varepsilon^d$ for every $q \in \mathbb{R}^{\geq 0}$, $n, d \in \mathbb{N}$ such that $n \geq n_0$ and $d \geq 2$. Fix such c and n_0 (these values will be specified explicitly at the end of the proof) and let $s \geq n_0$, let $n \in \mathbb{N}$ be a multiple of s and let $f_n := 4c(n/s-1)^{-d}$. Similarly to the proof of Proposition 5.7, for a set $\mathcal{S} \subseteq [n]$, let $H_{\mathcal{S}}$ denote the event that the j -th hit occurred for all $j \in \mathcal{S}$,¹⁹ and let $Q_{\mathcal{S}} \leftarrow \sum_{i \in \mathcal{S}} Q_i$ and $q_{\mathcal{S}} \leftarrow \mathbb{E}[Q_{\mathcal{S}}^d]$. We will prove that $\text{MH}_n(d, \varepsilon, q) \leq 2 \cdot q \cdot \varepsilon^d \cdot f_n$.

We first note that for every $j \in [n-s]$, $i \leftarrow j+s$:

$$\Pr[H_{(j,i)}] \leq c \cdot q_{(j,i)} \cdot \varepsilon^d \tag{7}$$

Otherwise, by simulating the first j calls made by $\mathbf{P}$, we get an algorithm that contradicts Proposition 5.7 (recall $s \geq n_0$). Next, if $c \cdot q_{(j,i)} \cdot \varepsilon^d \leq q \cdot \varepsilon^d \cdot f_n$, for some $j \in [n-s]$, then by Equation (7) we are done as $\text{MH}_n(d, \varepsilon, q) \leq \Pr[H_{(j,i)}]$. So, in the rest of the proof assume that $c \cdot q_{(j,i)} \cdot \varepsilon^d \geq q \cdot \varepsilon^d \cdot f_n$, and thus for all $j \in [n-s]$:

$$q_{(j,i)} \geq q \cdot f_n / c \tag{8}$$

In the following we further assume that for every $j \in (1, n/s]$: (we will address this assumption later)

$$Q_{[s(j-1)]} \geq \frac{j-1}{4} \cdot Q_{(s(j-1), sj)} \tag{9}$$

Since

$$\begin{aligned}
q &= \mathbb{E}[(\sum_{j \in [n/s]} Q_{((s(j-1), sj))})^d] \geq \sum_j \mathbb{E}[Q_{(s(j-1), sj)}^d + d \cdot Q_{((s(j-1), sj))} \cdot Q_{\leq s(j-1)}^{d-1}] \\
&> d \sum_{j \in [n/s]} \mathbb{E}[Q_{((s(j-1), sj))} \cdot Q_{\leq s(j-1)}^{d-1}].
\end{aligned} \tag{10}$$

¹⁹Obviously, $H_{\mathcal{S}}$ is implied by H_i for $i = \max(\mathcal{S})$, however it will be useful conceptually to consider all hits indexed by $\mathcal{S}$.

Equation (9) yields that

$$\begin{aligned}
q &> \frac{d}{4} \cdot \sum_{j \in [n/s]} (j-1)^{d-1} \cdot \mathbb{E}[Q_{(s(j-1),sj)}^d] = \frac{d}{4} \cdot \sum_{j \in [n/s]} (j-1)^{d-1} \cdot q_{(s(j-1),sj]} \\
&\geq \frac{d}{4} \cdot \sum_{j \in [n/s-1]} j^{d-1} \cdot q \cdot f_n/c \\
&\geq \frac{1}{4c} \cdot q \cdot f_n \cdot (n/s-1)^d.
\end{aligned} \tag{11}$$

The first inequality is by Equations (9) and (10), the second one by Equation (8), and the last one by Fact 5.3. We conclude that

$$\begin{aligned}
q &> \frac{1}{4c} \cdot q \cdot f_n \cdot (n/s-1)^d \iff \\
&4c(n/s-1)^{-d} > f_n.
\end{aligned} \tag{12}$$

Notice that Equation (12) contradicts our definition of f_n . Therefore Equation (8) fails for at least one $j \in [n-s]$, which concludes the proof for the case where Equation (9) holds true.

Removing the assumption of Equation (9). In the following, we argue that Equation (9) essentially does not incur any loss of generality.

Let $t \in \mathbb{N}$ for some $\varepsilon \in (0, 1]$ and $s \in \mathbb{N}$, and let $X_1, \dots, X_t$ be independent Boolean random variables, each takes the value 1 with probability ε . By Hoeffding's inequality:

$$\begin{cases} \Pr[\sum_{i \in [(j-1)s/2\varepsilon]} X_i \geq (j-1)s] \leq e^{-s(j-1)/6} \\ \Pr[\sum_{i \in [2s/\varepsilon]} X_i \leq s/2] \leq e^{-s/6} \end{cases} \tag{13}$$

Compute (assuming $i \geq 2$):

$$\begin{aligned}
&\Pr[Q_{(i-1)s,i} \geq 4Q_{\leq s(i-1)}/(i-1)] \\
&= \Pr[Q_{((i-1)s, is)} \geq 4Q_{\leq s(i-1)}/(i-1) \wedge H_{s(i-1)}] \\
&= \Pr[Q_{((i-1)s, is)} \geq 4Q_{\leq s(i-1)}/(i-1) \wedge H_{s(i-1)} \wedge Q_{\leq s(i-1)} \geq s(i-1)/2\varepsilon] \\
&\quad + \Pr[Q_{((i-1)s, is)} \geq 4Q_{\leq s(i-1)}/(i-1) \wedge H_{s(i-1)} \wedge Q_{\leq s(i-1)} < s(i-1)/2\varepsilon] \\
&\leq \Pr[Q_{((i-1)s, is)} \geq 2s/\varepsilon \wedge H_{(i-1)s}] + \Pr[H_{s(i-1)} \wedge Q_{\leq s(i-1)} < s(i-1)/2\varepsilon].
\end{aligned}$$

Therefore, by Equation (13),

$$\Pr[Q_{(i-1)s,i} \geq 4Q_{\leq s(i-1)}/(i-1)] \leq 2e^{-s/6} \tag{14}$$

Next, consider algorithm $\mathbf{P}_{\text{trim}}$, the variant of algorithm $\mathbf{P}$, that simply aborts whenever $Q_{(i-1)s,i} \geq 4Q_{\leq s(i-1)}/(i-1)$ for $i > 1$. By Equation (13) and union bound, the probability that $\mathbf{P}_{\text{trim}}$ makes n hits is at most $(n/s-1) \cdot 2 \cdot e^{-s/6} \leq q \cdot \varepsilon^d \cdot f_n$ smaller than that of $\mathbf{P}$, for suitable choice of s . Since Equation (9) holds for $\mathbf{P}_{\text{trim}}$, it follows that its success probability in getting n hits is at most $q_{\text{trim}} \cdot \varepsilon^d \cdot f_n$, for q_{trim} being the d^{th} -moment running time of $\mathbf{P}_{\text{trim}}$. It follows that $\mathbf{P}$'s success probability in getting n hits is at most $q \cdot \varepsilon^d \cdot f_n + (n/s-1) \cdot 2 \cdot e^{-s/6} \leq 2 \cdot q \cdot \varepsilon^d \cdot f_n$ for suitable choice of parameters (specified below).

Deducing explicit parameters. It remains to specify the parameters that we left undefined so far. Namely, c , n_0 and the minimal value of s . For s , we require that $(n/s - 1) \cdot 2 \cdot e^{-s/6} \leq q \cdot \varepsilon^d \cdot f_n$ which is true under the sufficient condition that $s \geq 6(d+1) \ln(n/\varepsilon)$. Therefore, taking the smallest possible s that is a power of two, we deduce that $s \leq 12(d+1) \ln(n/\varepsilon)$ and therefore, taking n to be a power of two as well, we require that $n \geq 24(d+1) \ln(n/\varepsilon)$.

Next, setting $c = 2$, it suffices that $2 \geq \varepsilon^{(1-d) \cdot (1-1/d)^n}$ which is always true when $n \geq d \ln(1/\varepsilon)$. In conclusion, as long as $2^\ell \geq 24(d+1) \ln(2^\ell/\varepsilon) \geq d \ln(1/\varepsilon)$, it holds that

$$\begin{aligned} \text{MH}_{2^\ell}(d, \varepsilon, q) &\leq 2 \cdot q \cdot \varepsilon^d \cdot 4c \cdot \left(\frac{1}{\frac{2^\ell}{12(d+1) \ln(2^\ell/\varepsilon)} - 1} \right)^d \\ &\leq q \cdot \varepsilon^d \cdot 16 \cdot \left(\frac{24(d+1) \cdot \ln(2^\ell/\varepsilon)}{2^\ell} \right)^d. \end{aligned}$$

□

5.3 Lower Bounds on the Multi-Hitting Games

In this section, we prove that our upper bounds on hitting games are essentially tight. Specifically, we show that Theorem 5.6 is tight up to a constant factor when $n = 2$ (the case of interest for bounding the security of identification schemes), and tight up to logarithmic factors in the general case.

5.3.1 Case $n = 2$

In this section, we show that there exists an algorithm who wins the double-hitting game with probability very close to $\delta \leftarrow (\mathbb{E}[Q^d]^{1/d} \varepsilon)^{2-1/d}$ (specifically at least $\delta/20$), where $\mathbb{E}[Q^d]^{1/d}$ can take essentially any arbitrary value.²⁰

Theorem 5.9 (Lower bounds on the double-hitting games). *Let $d \geq 2$ and $\varepsilon \in [0, 1]$. Then for any $q_0 \in [[1/16\varepsilon]]$ there exists an algorithm P for the double-hitting game such that $\mathbb{E}[Q^d]^{1/d} \geq q_0$ and $\Pr[W] \geq (\mathbb{E}[Q^d]^{1/d} \cdot \varepsilon)^{2-1/d}/20$, where Q and W denote, respectively, the number of queries and the winning event of P in Game 5.4 with $n = 2$.*

We prove Theorem 5.9 by defining an algorithm for the multiple-hitting game and analyzing its query complexity and success probability.

Algorithm 5.10. $[P_{a,b}]$

Parameters: $a, b \in \mathbb{N}$.

Operation:

1. Make a queries. Abort if no hit occurs.
2. Make a total of $b \geq a$ queries. Halt if a second hit occurs.

²⁰The case of worst-case query complexity is straightforward, and it is omitted from the text.

Proposition 5.11 (Running time and success probability of Algorithm 5.10). *Let $a, b, d \in \mathbb{N}$, let $\varepsilon \in [0, 1]$, let $P = P_{a,b}$ be according to Algorithm 5.10 and let $G = G_{2,\varepsilon}$ be according to Game 5.4. Let Q and W denote, respectively, the number of queries and the winning event of P in G . Then, $E[Q^d] \leq a^d + a\varepsilon b^d$. Furthermore, if $b \cdot \varepsilon \leq 1/2$, then $E[Q^d] \geq 1/2 \cdot (a^d + \varepsilon ab^d)$ and $\Pr[W] \geq ab\varepsilon^2/4$.*

Proof. The bound $E[Q^d] \leq a^d + a\varepsilon b^d$ is immediate by noting that there are exactly a possible hits and the maximal number of queries when a hit occurs is b . The second bound $E[Q^d] \geq 1/2 \cdot (a^d + \varepsilon ab^d)$ is straightforward as $1/2 \leq (1 - b\varepsilon) \leq (1 - \varepsilon)^{b-1} \leq (1 - \varepsilon)^a$. For the third bound, $\Pr[W] \geq ab\varepsilon^2/4$, observe that

$$\begin{aligned} \Pr[W] &= 1 - (1 - \varepsilon)^a - a\varepsilon(1 - \varepsilon)^{b-1} \\ &\geq 1 - (1 - a\varepsilon + a(a-1)\varepsilon^2/2) - a\varepsilon(1 - (b-1)\varepsilon + (b-1)(b-2)\varepsilon^2/2) \end{aligned} \quad (15)$$

$$\begin{aligned} &= a(b-a/2)\varepsilon^2 - a(b-1)(b-2)\varepsilon^3/2 \\ &\geq ab\varepsilon^2/2 - ab^2\varepsilon^3/2 \\ &\geq ab\varepsilon^2/2 - ab\varepsilon^2/4 \\ &\geq ab\varepsilon^2/4. \end{aligned} \quad (16)$$

Equation (15) follows from Fact 5.2 and Equation (16) follows from the fact that $b\varepsilon \leq 1/2$. $\square$

Given Proposition 5.11, we are ready to prove Theorem 5.9.

Proof of Theorem 5.9. For $a \leq \varepsilon^{-1}/16$, to be determined by the analysis, set $b \leftarrow \lceil a^{1-1/d}/\varepsilon^{1/d} \rceil$. Let $P = P_{a,b}$ be according to Algorithm 5.10. Since

$$b\varepsilon = \lceil a^{1-1/d}/\varepsilon^{1/d} \rceil \cdot \varepsilon \leq 2 \cdot (a^{1-1/d}/\varepsilon^{1/d}) \cdot \varepsilon \leq 2 \cdot (a\varepsilon)^{1-1/d} \leq 2 \cdot (1/16)^{1-1/d} \leq 1/2,$$

Proposition 5.11 yields that $E[Q^d] \leq a^d + a\varepsilon b^d$ and $\Pr[W] \geq ab\varepsilon^2/4$. We conclude the proof by bounding $a^d + a\varepsilon b^d$ and $ab\varepsilon^2/4$. We first note that

$$\begin{aligned} a^d + a\varepsilon(a^{1-1/d}/\varepsilon^{1/d})^d &\leq a^d + a\varepsilon b^d \leq a^d + a\varepsilon(a^{1-1/d}/\varepsilon^{1/d})^d \cdot 2^d && \iff \\ 2a^d &\leq a^d + a\varepsilon b^d \leq a^d \cdot (1 + 2^d) \end{aligned} \quad (17)$$

Hence, $a \leq E[Q^d]^{1/d} \leq a \cdot \sqrt{5}$ for every $d \geq 2$. We also note that

$$\begin{aligned} ab\varepsilon^2/4 &\geq a(a^{1-1/d}/\varepsilon^{1/d})\varepsilon^2/4 \\ &= (a\varepsilon)^{2-1/d}/4 \\ &\geq (E[Q^d]^{1/d}/\sqrt{5} \cdot \varepsilon)^{2-1/d}/4 \\ &\geq (E[Q^d]^{1/d} \cdot \varepsilon)^{2-1/d}/20. \end{aligned}$$

Hence, $\Pr[W] \geq (E[Q^d]^{1/d} \cdot \varepsilon)^{2-1/d}/20$. The proof now follows by taking $a \leftarrow q_0$. $\square$

5.3.2 Arbitrary n

In this section, we show that there exists an algorithm with winning probability $q \cdot (\varepsilon/n)^d$, which is only a poly-logarithmic factor away from the bound in Theorem 5.6 (where the polylogarithmic factor depends on n/ε).

Theorem 5.12 (Lower bounds on the multiple-hitting games). *Let $d \geq 2$ and $\varepsilon \in [0, 1]$. Then for any $q_0 \in (0, (n/\varepsilon)^d)$ there exists an algorithm $\mathbf{P}$ for the n -hitting game such that $\mathbb{E}[Q^d] \geq q_0$ and $\Pr[W] \geq \mathbb{E}[Q^d] \cdot (\varepsilon/16n)^d$, where Q and W denote, respectively, the number of queries and the winning event of $\mathbf{P}$ in Game 5.4.*

We prove Theorem 5.12 by defining an algorithm for the multiple-hitting game and analyzing its query complexity and success probability.

Algorithm 5.13. $[\mathbf{P}_{\ell, \varepsilon, n}]$

Parameters: $n \in \mathbb{N}$, $\varepsilon \in [0, 1]$ and $\ell \in (0, (n/\varepsilon)^d)$.

Operation:

1. Toss a coin with probability $\ell/(n/\varepsilon)^d$. Abort if coin returns zero.
2. Make a total of $\lceil 4n/\varepsilon \rceil$ queries. Halt if an n -th hit occurs.

Theorem 5.12 is a corollary of the proposition below.

Proposition 5.14 (Running time and success probability of Algorithm 5.13). *Let $a, b, d \in \mathbb{N}$, let $\varepsilon \in [0, 1]$, let $\mathbf{P} = \mathbf{P}_{\ell, \varepsilon, n}$ be according to Algorithm 5.13 and let $\mathbf{G} = \mathbf{G}_{n, \varepsilon}$ be according to Game 5.4. Let Q and W denote, respectively, the number of queries and the winning event of $\mathbf{P}$ in $\mathbf{G}$. Then, $\mathbb{E}[Q^d] \in (\ell/8^d, 8^d \ell)$. Furthermore, $\Pr[W] \geq \mathbb{E}[Q^d] \cdot (\varepsilon/16n)^d$.*

Proof. Let B denote the outcome of the coin in the first step of Algorithm 5.13. Clearly

$$\begin{aligned} \mathbb{E}[Q^d] &= \frac{\ell}{(n/\varepsilon)^d} \lceil 4(n/\varepsilon) \rceil^d \\ &\leq \frac{\ell}{(n/\varepsilon)^d} \cdot (8(n/\varepsilon))^d \leq 8^d \ell \end{aligned} \tag{18}$$

Next, by Chernoff, observe that $\Pr[W \mid B = 1] \geq 1/2$ and $\Pr[Q \leq n/(4\varepsilon) \wedge W \mid B = 1] \leq 1/2$. We will use these inequalities to lower-bound $\mathbb{E}[Q^d]$ and $\Pr[W]$. First, we compute

$$\begin{aligned} \mathbb{E}[Q^d] &\geq \Pr[Q \geq n/(4\varepsilon) \wedge B = 1] \cdot \mathbb{E}[Q^d \mid Q \geq n/(4\varepsilon) \wedge B = 1] \\ &\geq \Pr[Q \geq n/(4\varepsilon) \mid B = 1] \cdot \Pr[B = 1] \cdot \left(\frac{n}{4\varepsilon}\right)^d \\ &\geq \Pr[Q \geq n/(4\varepsilon) \mid B = 1] \cdot \frac{\ell}{(n/\varepsilon)^d} \cdot \left(\frac{n}{4\varepsilon}\right)^d \\ &= (1 - \Pr[Q < n/(4\varepsilon) \wedge W \mid B = 1]) \cdot \frac{\ell}{(n/\varepsilon)^d} \cdot \left(\frac{n}{4\varepsilon}\right)^d \\ &\geq \frac{1}{2} \cdot \frac{\ell}{(n/\varepsilon)^d} \cdot \left(\frac{n}{4\varepsilon}\right)^d \geq \frac{\ell}{2 \cdot 4^d} \end{aligned} \tag{19}$$

Finally,

$$\begin{aligned}
\Pr[W] &= \Pr[W \mid B] \cdot \Pr[B] \geq \frac{1}{2} \cdot \frac{\ell}{(n/\varepsilon)^d} \\
&\geq \frac{\mathbb{E}[Q^d]}{8^d \cdot \ell} \cdot \frac{1}{2} \cdot \frac{\ell}{(n/\varepsilon)^d} \\
&\geq \mathbb{E}[Q^d] \cdot \left(\frac{n}{16\varepsilon}\right)^d
\end{aligned} \tag{20}$$

□

5.4 The Coins and Cows Game

For convenience, we recall the coins and cows game.

Game 5.15. [Coins and cows]

Parameters: $\varepsilon \in [0, 1]$ and $b, s \in \mathbb{N}$.

Parties: Challenger C and player P.

The game:

1. C: sets $w \leftarrow 0$.
2. On each query made by P, C acts as follows:
 - (a) Set $c' \leftarrow 0$ and sample $c \stackrel{\text{R}}{\leftarrow} \text{Ber}_\varepsilon$.
 - (b) If $c = 1$:
 - i. If $w \geq 1$: sample $c' \stackrel{\text{R}}{\leftarrow} \text{Ber}_{\varepsilon'}$ for $\varepsilon' \leftarrow \frac{1}{b} + \frac{w}{s}$.^a
 - ii. Set $w \leftarrow w + 1$.
 - (c) Send (c, c') to P.
3. P *wins* if at some point it holds that $c' = 1$.

^aWe use the convention that Ber_δ for $\delta > 1$ equals Ber_1 , i.e., the constant one distribution.

Theorem 5.16 (Upper bounds on the coins-and-cows game,²¹ restating Theorem 4.10). *For $\varepsilon \in [0, 1]$, $b, s \in \mathbb{N}$ and $q \in \mathbb{R}^+$:*

Worst-case query complexity.

$$\text{RDH}(\varepsilon, b, s, q) \leq \begin{cases} q^2 \cdot \varepsilon^2 \cdot (1/b + q/s) & \text{if } q\varepsilon < 1 \\ q \cdot \varepsilon \cdot (1/b + q/s) & \text{otherwise} \end{cases}.$$

²¹The term q/s in the bound can be significantly reduced for most choices of q and ε , but we state the bound as above since, in our application, s dominates the other parameters.

Higher-moment query complexity. For $d \geq 2$:

$$\text{RDH}(d, \varepsilon, b, s, q) \leq \begin{cases} \frac{2u(1+ub/s)}{b} \cdot (q^{1/d} \cdot \varepsilon)^{2-1/d} & \text{if } q^{1/d} \cdot \varepsilon < 1 \\ \frac{2u(1+ub/s)}{b} \cdot q\varepsilon^d & \text{otherwise,} \end{cases}$$

letting $u \leftarrow 16 \cdot (48 \cdot (d+1) \cdot \ln((s+1)/\varepsilon))^d$.

Proof.

Worst-case bound. The worst case bound follows by a straightforward union bound by taking the largest possible probability $1/b + q/s$ for the coin c' .

Higher-moment bound. Fix $\varepsilon > 0$, $d \geq 2$, and a player P of d^{th} -moment query complexity q . Define the following random variables and events: let I denote the winning index over the indices satisfying $c = 1$ in the coins and cows game (set to 0 if no such index exists), and let H_n denote the event that n hits occurred. We use the following quantities: let $a_n \leftarrow \frac{1}{b} + \frac{n-1}{s}$, let $u_n \leftarrow 16 \cdot (48 \cdot (d+1) \cdot \ln(n/\varepsilon))^d$, and let n_0 be the smallest n satisfying $n \geq u_n$. Finally, let $s_0 = \min(n_0 - 1, s + 1)$. Observe that

$$\Pr[I = n] \leq a_n \cdot \Pr[H_n] \leq \begin{cases} a_n \cdot (q^{1/d} \varepsilon)^{2-1/d} & \text{if } n < n_0 \\ a_n \cdot u_n \cdot q \cdot (\varepsilon/n)^d & \text{otherwise.} \end{cases} \quad (21)$$

The above holds by Theorem 5.6 where for the case $n < n_0$ we additionally use the fact that $\Pr[H_n] \leq \Pr[H_2]$. Therefore:

$$\begin{aligned} \text{RDH}(d, \varepsilon, b, s, q) &= \sum_{n=2}^{s_0} \Pr[I = n] + \sum_{n \geq s_0+1} \Pr[I = n] \\ &\leq \frac{(q^{1/d} \varepsilon)^{2-1/d}}{b} \cdot \sum_{n=2}^{s_0} a_n b + \frac{q \cdot \varepsilon^d}{b} \cdot \sum_{n=s_0+1}^{s+1} b a_n u_n \cdot \frac{1}{n^d} \\ &\leq \frac{(q^{1/d} \varepsilon)^{2-1/d}}{b} \cdot \left(s_0 + b \cdot \frac{s_0^2}{s} \right) + \frac{q \cdot \varepsilon^d}{b} \cdot u_{s+1} \cdot \sum_{n=s_0+1}^{s+1} b a_n \cdot \frac{1}{n^d} \\ &\leq \frac{(q^{1/d} \varepsilon)^{2-1/d}}{b} \cdot \left(s_0 + b \cdot \frac{s_0^2}{s} \right) + \frac{q \cdot \varepsilon^d}{b} \cdot u_{s+1} \cdot \left(1 + \frac{b}{s} \cdot \sum_{n=s_0+1}^{s+1} \frac{1}{n^{d-1}} \right) \\ &\leq \frac{(q^{1/d} \varepsilon)^{2-1/d}}{b} \cdot \left(s_0 + b \cdot \frac{s_0^2}{s} \right) + \frac{q \cdot \varepsilon^d}{b} \cdot u_{s+1} \cdot \left(1 + \frac{b(1 + \ln(s+1))}{s} \right). \end{aligned} \quad (22)$$

The first inequality is by Equation (21). The second and third inequalities uses the fact that $u_1, \dots, u_n$ is an increasing function of n , as well as the standard fact $\sum_{i=1}^k i \leq k^2$ and $\sum_{i=1}^k 1/i^2 \leq 1$. The last inequality uses the fact that $\sum_{i=1}^k 1/i \leq 1 + \ln(k)$. We conclude that

$$\begin{aligned} \text{RDH}(d, \varepsilon, b, s, q) &\leq \frac{(q^{1/d} \varepsilon)^{2-1/d}}{b} \cdot \left(u_{s+1} + b \cdot \frac{u_{s+1}^2}{s} \right) + \frac{q \cdot \varepsilon^d}{b} \cdot \left(1 + \frac{u_{s+1} b (1 + \ln(s+1))}{s} \right) \\ &\leq \begin{cases} \frac{(q^{1/d} \varepsilon)^{2-1/d}}{b} \cdot 2 \left(u_{s+1} + b \cdot \frac{u_{s+1}^2}{s} \right) & \text{if } q^{1/d} \varepsilon < 1 \\ \frac{q \cdot \varepsilon^d}{b} \cdot 2 \left(u_{s+1} + b \cdot \frac{u_{s+1}^2}{s} \right) & \text{otherwise,} \end{cases} \end{aligned}$$

The first inequality follows by observing that $u_{s+1} \geq 2 + \ln(s+1)$, and that $s_0 \leq u_{s+1}$ as either $s_0 = n_0 - 1$ and thus $n_0 - 1 \leq u_{n_0-1} \leq u_{s+1}$, or $s_0 = s + 1$ and thus $s_0 = s + 1 \leq u_{s+1}$. The second inequality is a simple case analysis. $\square$

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